Supplement to "Deep Regression for repeated measurements"

In this supplementary material, we provide the additional results, auxiliary lemmas, and all proofs for the main manuscript. Section S.1 further elucidates more results on circumventing the curse of dimensionality from Section 4. In Section S.2, we extend the discussion on regression splines, a supplementing result of Section 3. Additional numerical results are shown in Section S.3. Then Section S.4 presents a collection of technical lemmas. The detailed proofs of theorems and corollaries can be found in Section S.5 and proofs of the the lemmas are given in Section S.6.

S.1 Additional Results on Circumventing Curse of Dimensionality

S.1.1 Suboptimality of Linear Estimators for Hierarchical Composition Structure

Schmidt-Hieber (2020) demonstrated that wavelet series estimators are not adaptive to the unknown composition structure in the regression function, even for the single index model. Suzuki & Nitanda (2021) then extended to the multi-index model and a broader class of linear estimators. Here, the linear estimators for cross-sectional regression are any estimators with the form

$$\tilde{f}_L(x) = \sum_{i=1}^{n} Y_i \varphi_i \left(x, X_1, X_2, \ldots, X_n\right),$$

where $\varphi_i, i = 1, 2..., n$ are measurable functions that only depend on $x, X_1, X_2, \ldots, X_n$. It is

easy to verify that the linear estimator class includes local polynomials, RKHS, regression splines, wavelet series and so on. Similarly, one can show, in the repeated measurements setting, the linear estimators

$$\tilde{f}_L(x) = \sum_{i=1}^{n}\sum_{j=1}^{m} Y_{ij}\varphi_{ij}\left(x, X_{11}, \ldots, X_{nm}\right),$$

are also not adaptive to unknown composition structures. For simplicity, we only discuss the single index models.

Theorem S.1. *Consider the repeated measurement model (1) and assume that $\mathcal{P}_X$ has a uniform distribution on $\Omega = [0,1]^d$, $d \geq 3$. Then we have*

$$\inf_{\tilde{f}_L} \sup_{\mathcal{P}_{Z,\epsilon} \in \mathcal{Q}_{Z,\epsilon}} \mathcal{E}_{f^\circ}(\tilde{f}_L) \geq c_{20}\left(n^{-1} + (nm)^{-(2s+d)/(2s+2d)}\right),$$

where the infimum is taken over all linear estimators $\tilde{f}_L$ based on the available data set $\{(Y_{ij}, X_{ij}) : 1 \leq i \leq n, 1 \leq j \leq m\}$ with form $\tilde{f}_L(x) = \sum_{i=1}^{n}\sum_{j=1}^{m} Y_{ij}\varphi_{ij}\left(x, X_{11}, \ldots, X_{nm}\right)$ and the supremum is taken over all distributions $\mathcal{P}_{Z,\epsilon} \in \mathcal{Q}_{Z,\epsilon}$ satisfying Assumption 2 and $f^\circ(x) = g(A^T x + b)$ for some $\|g\|_{\mathcal{C}^s([0,1])} \leq 1$, $A \in \mathbb{R}^d, b \in \mathbb{R}$ s.t. $Ax + b \in [0,1]$, $\forall x \in [0,1]^d$, and c_{20} is a positive number not depending on n and m.

Theorem S.1 provides a lower bound that the convergence rate of linear estimators cannot be faster than $n^{-1} + (nm)^{-(2s+d)/(2s+2d)}$, which is slower than the optimal rate $n^{-1}+(nm)^{-2s/(2s+1)}$ achieved by the DNN estimator given in Theorem 4.2 when $2s+d < 2sd$. This illustrates that the linear estimators are not adaptive to the hierarchical composition structure, i.e., the minimax rate cannot be achieved when the structure is unknown. Therefore, traditional methods often assume a specific structure of the real model and then design the tailored estimation method for that known structure. In contrast, the DNNs can adapt to unknown low intrinsic dimension structures through optimization because the minimizer can learn the unknown hierarchical structures without specific knowledge and designs.

S.1.2 Anisotropic Hölder Smoothness

In the previous subsection, we have studied that the function of interest obeys a general presumed structural assumption and has shown success in theoretical terms. However, assuming a hierarchical composition structure for each regression problem is impractical. To confront this challenge, another effective approach is to consider the functions that exhibit varying smoothness along different coordinate directions, as improving smoothness isotropically can be a daunting task (Bhattacharya et al. 2014). This is also practically sensible, given that the coordinates may have distinct meanings, such as spatial and temporal features, thus different smoothness is often preferred for different coordinates. Based on these considerations, we introduce the definition of anisotropic Hölder space as follows.

Definition S.1. *Let* $\boldsymbol{s} = (s_1, s_2, \ldots, s_d), s_i > 0$ *be the smoothness index. Assume* $d \geq 2$*, the anisotropic* $\boldsymbol{s}$*-Hölder norm of a function* $f : \Sigma \in \mathbb{R}^d \to \mathbb{R}$ *is defined as*

$$\|f\|_{\mathcal{C}^{\boldsymbol{s}}} = \sum_{i=1}^{d} \sum_{\alpha < s_i} \|\partial_i^{\alpha} f\|_{\infty} + \sum_{i=1}^{d} \sum_{\alpha = \lfloor s_i \rfloor} \sup_{\substack{x, x' \in \Sigma \\ x' = x + te_i, t \neq 0}} \frac{|\partial_i^{\alpha} f(x) - \partial_i^{\alpha} f(x')|}{\|x - x'\|^{s_i - \lfloor s_i \rfloor}},$$

where ∂_i^{α} *is the* α*-th order partial derivative of* f *with respect to the* i*-th component* e_i *with* $\alpha \in \mathbb{N}$*. Furthermore, the anisotropic Hölder space* $\mathcal{C}^{\boldsymbol{s}}(\Sigma)$ *consists of all functions for which the anisotropic* $\boldsymbol{s}$*-Hölder norm is finite.*

Some existing works consider similar anisotropic smoothness functions for cross-sectional regression. Hang & Steinwart (2021) investigated the nonparametric regression problem using SVMs with anisotropic Gaussian RBF kernels and established the optimal convergence rates. They assume knowing function smoothness in each direction to establish the theory, and their method needs to tune $(d+1)$ parameters. Recently, Suzuki & Nitanda (2021) developed nonparametric regression theory via sparsely connected DNNs. From the previous literature, the optimal rate for cross-sectional regression is $O(n^{-2\bar{s}/(2\bar{s}+d)})$, where

$\tilde{s}$ is the harmonic mean of the smoothness in each direction

$$\tilde{s} := \left(\frac{1}{d} \sum_{i=1}^{d} \frac{1}{s_i} \right)^{-1}.$$

Now, we first build the approximation theorem of fully connected DNNs for anisotropic Hölder function.

Theorem S.2 (Approximation for anisotropic Hölder function)**.** *Suppose that $f \in \mathcal{C}^{\boldsymbol{s}}(\Omega)$ with the anisotropic $\boldsymbol{s}$-Hölder norm $\|f\|_{\mathcal{C}^{\boldsymbol{s}}} \leq B_5$ for some $B_5 > 0$ and $\|f\|_\infty \leq B_1$. Then, for any $L, W \geq 3$, there is some neural network $f_{\mathcal{FNN}} \in \mathcal{FNN}(d, c_{21} L \log L, c_{22} W \log W, B_1)$ satisfying*

$$|f_{\mathcal{FNN}}(x) - f(x)| \leq c_{23} (LW)^{-2\tilde{s}/d},$$

for each $x \in \Omega$, where c_{21} and c_{22} are constants not depending on L, W and f, and c_{23} is a constant free of L and W.

By assuming the anisotropic Hölder smoothness, we obtain the non-asymptotic upper bound and the matching minimax lower bound that lead to the optimal convergence rate for the repeated measurements model.

Assumption S.1 (Anisotropic Hölder smoothness)**.** *The target function $f^\circ(x)$ belongs to the anisotropic Hölder space $\mathcal{C}^{\boldsymbol{s}}(\Omega)$ for a given positive $\boldsymbol{s}$ and $\|f^\circ\|_{\mathcal{C}^{\boldsymbol{s}}} \leq B_5$.*

Theorem S.3. *Consider model (1) and the estimator $\hat{f}_{nm}$ obtained by (2). Suppose that Assumptions 1, 2 and S.1 hold, and set $\mathcal{F}_{nm} = \mathcal{FNN}(d, c_{21} L \log L, c_{22} W \log W, B_1)$. If we specify the neural network in a flexible way satisfying $LW = \lfloor c_{24} (nm)^{d/(4\tilde{s}+2d)} (\log nm)^{-4d/(2\tilde{s}+d)} \rfloor$ for some certain constant c_{24} not depending on n and m, then*

$$\mathcal{E}_{f^\circ}(\hat{f}_{nm}) \leq c_{25} (n^{-1} + (nm)^{-2\tilde{s}/(2\tilde{s}+d)} (\log nm)^{16\tilde{s}/(2\tilde{s}+d)}),$$

where c_{25} is a constant free of n and m.

Theorem S.4. *Consider the repeated measures model (1) and assume that $\mathcal{P}_X$ has a uniform distribution on Ω. Under the Assumptions 2 and S.1, we have*

$$\inf_{\tilde{f}} \sup_{\mathcal{P}_{Z,\epsilon} \in \mathcal{Q}_{Z,\epsilon}} \mathcal{E}_{f^\circ}(\tilde{f}) \geq c_{26}(n^{-1} + (nm)^{-2\tilde{s}/(2\tilde{s}+d)}),$$

where the infimum is taken over all possible estimators $\tilde{f}$ based on the available data set $\{(Y_{ij}, X_{ij}) : 1 \leq i \leq n, 1 \leq j \leq m\}$, the supremum is taken over all distributions $\mathcal{P}_{Z,\epsilon} \in \mathcal{Q}_{Z,\epsilon}$ satisfying Assumption 2 and S.1, and c_{26} is a positive number not depending on n and m.

Similar conclusions can be drawn from these theorems that our ReLU DNN estimator attains the minimax optimal rate $O(n^{-1} + (nm)^{-2\tilde{s}/(2\tilde{s}+d)})$ up to logarithmic factors for the repeated measurement model (1), with the phase transition occurs at $m \asymp O(n^{-d/2\tilde{s}})$. It is worth noting that in the anisotropic framework $f^\circ \in \mathcal{C}^{\boldsymbol{s}}$ in contrast to $f^\circ \in \mathcal{C}^{s_{\min}}$ in the isotropic case with the convergence rate $O(n^{-1} + (nm)^{-2s_{\min}/(2s_{\min}+d)})$ by Theorem 3.1. Knowing $\min_{1\leq i\leq d} s_i \leq \tilde{s} \leq \max_{1\leq i\leq d} s_i$, the harmonic mean $\tilde{s}$ can be significantly larger than $s_{\min}$ when the smoothness is highly anisotropic.

S.1.3 Low-dimensional Support

For completeness of the paper, we first briefly introduce some of the basic concepts about manifolds used.

Definition S.2 (Chart)**.** *A chart for $\mathcal{M}$ is a pair (U, ϕ) such that $U \subset \mathcal{M}$ is open and $\phi : U \to \mathbb{R}^d$, where ϕ is a homeomorphism, i.e. ϕ is a bijective, both ϕ and the inverse of ϕ are continuous.*

Definition S.3 (C^k compatible)**.** *We say two charts (U, ϕ) and (V, ψ) on $\mathcal{M}$ are C^k compatible if and only if*

$$\phi \circ \psi^{-1} : \psi(U \cap V) \mapsto \phi(U \cap V) \quad \text{and} \quad \psi \circ \phi^{-1} : \phi(U \cap V) \mapsto \psi(U \cap V)$$

are both C^k.

Definition S.4 (Atlas). *An atlas for $\mathcal{M}$ is a collection $\{(U_i, \phi_i)\}_{i\in I}$ of charts such that $\bigcup_i U_i = \mathcal{M}$. Then a C^k atlas for $\mathcal{M}$ is an atlas consisting of pairwise C^k compatible charts.*

Definition S.5 (Smooth manifold and Riemannian manifold). *A smooth manifold is a manifold together with a C^∞ atlas. A Riemannian manifold is a smooth manifold $\mathcal{M}$ endowed with Riemannian metric.*

Definition S.6 (Hölder space $\mathcal{C}^s(\mathcal{M})$). *Let $\mathcal{M}$ be a $d_{\mathcal{M}}$-dimensional Riemannian manifold isometrically embedded in $\mathbb{R}^d$. Let $\{(U_i, \phi_i)\}_{i\in\mathcal{I}}$ be a C^∞ atlas of $\mathcal{M}$ where the $\phi_i's$ are orthogonal projections onto tangent space. Let s be a positive number. We say $f \in \mathcal{C}^s(\mathcal{M})$ if and only if, for any chart (U_i, ϕ_i), the composition $f \circ \phi^{-1} : \phi(U) \mapsto \mathbb{R}$ satisfies $f \circ \phi^{-1} \in \mathcal{C}^s(\phi(U))$.*

It is important to note that the above definition does not depend on the choice of the atlas $\{(U_i, \phi_i)\}_{i\in\mathcal{I}}$. Given chart (U, ϕ), consider an alternate chart (V, ψ) such that the intersection $V \cap U$ is not empty. In this context, the function $f \circ \psi^{-1}$ can be decomposed as $(f \circ \phi^{-1}) \circ (\phi \circ \psi^{-1})$. Because $f \circ \phi^{-1}$ is s-Hölder smooth and $\phi \circ \psi^{-1}$ is C^∞ smooth, their composition is still $\mathcal{C}^s$ smooth.

We also extend to the approximate manifold and Minkowski dimension set, coupling with the latest approximation theory of DNN estimators based on fully connected ReLU neural networks (Jiao et al. 2023). We keep Assumption 3 that $f^\circ \in \mathcal{C}^s\left([0,1]^d\right)$. Based on it, we assume that the predictor X is supported on some lower dimensional sets.

Let $\mathcal{M}$ be a compact $d_{\mathcal{M}}$-dimensional Riemannian submanifold and define the approximate manifold as

$$\mathcal{M}_\rho = \left\{x \in [0,1]^d : \inf\left\{\|x - y\|_2 : y \in \mathcal{M}\right\} \le \rho\right\}, \rho \in (0,1).$$

Since, in practice, observations may not fall strictly on a manifold but still tend to have a low-dimensional structure, as the empirical studies in Carlsson (2009) reveal for image data. Combining Corollary 2.2 and the approximation theory in the proof of Theorem 6.1 in Jiao et al. (2023), we have the following result.

Assumption S.2 (Approximate manifold)**.** *Assume the predictor X is supported on an approximate manifold $\mathcal{M}_\rho$.*

Theorem S.5. *Consider the repeated measures model (1) and the estimator $\hat{f}_{nm}$ obtained by (2). Suppose that Assumptions 1, 2, 3 and S.2 hold, and set the candidate class*

$$\mathcal{F}_{nm} = \mathcal{FNN}\left(d, c_{27} L \log L, c_{28} W \log W, B_1\right).$$

with c_{27}, c_{28} not depending on L, W. If we specify the neural network in a flexible way satisfying

$$LW = \lfloor c_{29}(nm)^{d_\delta/(4s+2d_0)}(\log nm)^{4d_\delta/(2s+d_\delta)} \rfloor$$

with some constant c_{29} not depending on m and n, and $d_\delta = O\left(d_{\mathcal{M}} \log(d/\delta)/\delta^2\right)$ is an integer such that $d_{\mathcal{M}} \leq d_\delta < d$ for any $\delta \in (0,1)$, then

$$\mathcal{E}_{f^\circ}(\hat{f}_{nm}) \leq c_{30}(n^{-1} + (nm)^{-2s/(2s+d_\delta)}(\log nm)^{16s/(2s+d_\delta)}),$$

where c_{30} is a constant free of n and m.

Another effective method to measure the intrinsic complexity of a set is the Minkowski dimension. Given a set $\Sigma \subseteq \mathbb{R}^d$, define the Minkowski dimension as

$$\mathrm{Mdim}(\Sigma) := \lim_{\varepsilon \to 0} \frac{\log \mathcal{N}\left(\varepsilon, \Sigma, L^2\right)}{\log(1/\varepsilon)},$$

if the limit on the right side exists. The Minkowski dimension characterizes the limiting behavior of the covering number when the radius of the balls converges to zero, which

provides an intuitive measure for the complexity of a set. It is worth noting that the Minkowski dimension does not depend on smoothness, thus it can measure the intrinsic dimension of some highly non-smooth sets.

Assumption S.3 (Minkowski dimension)**.** *The predictor X is supported on a set $A \subset \Omega$ with Minkowski dimension d^* less than d.*

Theorem S.6. *Consider the repeated measures model (1) and the estimator $\hat{f}_{nm}$ obtained by (2). Suppose that Assumptions 1, 2, 3 and S.3 hold, and set the candidate class*

$$\mathcal{F}_{nm} = \mathcal{FNN}\left(d, c_{31} L \log L, c_{32} W \log W, B_1\right).$$

with c_{31}, c_{32} not depending on L, W. If we specify the neural network in a flexible way satisfying

$$LW = \lfloor c_{33} (nm)^{d_0/(4s+2d_0)} (\log nm)^{4d_0/(2s+d_0)} \rfloor$$

with some constant c_{33} not depending on m and n, and $d \geq d_0 \geq \kappa d^*/\delta^2 = O\left(d^*/\delta^2\right)$ for $\delta \in (0,1)$ and some constant $\kappa > 0$, then

$$\mathcal{E}_{f^\circ}(\hat{f}_{nm}) \leq c_{34} (n^{-1} + (nm)^{-2s/(2s+d_0)} (\log nm)^{16s/(2s+d_0)}),$$

where c_{34} is a constant free of n and m.

Note that the convergence rate depends on $d_\delta = O(d_{\mathcal{M}} \log(d))$ under the approximate manifold assumption and $d_0 = O\left(d^*/\delta^2\right)$ under the low Minkowski dimension assumption, respectively, which are smaller than d but still greater than the intrinsic dimension. These error bounds may be sharpened given improved approximation theory that remains an open problem in theoretical deep learning.

S.2 Results on Regression Splines

We present here a typical regression spline formulation in the form of the cardinal B-splines, which is also used extensively in later proofs. Let the univariate function $B(x) = I(x \in [0,1])$ be the indicator function on for the interval $[0,1]$, and the cardinal B-splines of order r are defined recursively by

$$B_1(x) = B(x), \quad B_r(x) = B_{r-1}(r) * B(x),$$

where $*$ denotes convolution, i.e. $f * g(x) := \int f(x-t)g(t)\mathrm{d}t$. Then an explicit expression of $B_r(x)$ is given by

$$B_r(x) = \frac{1}{(r-1)!}\sum_{i=0}^{r}(-1)^i\binom{r}{i}(x-i)_+^{r-1}, \tag{S.1}$$

which is a piecewise polynomial of degree $(r-1)$ with the support $[0,r]$ and $\|B_r\|_\infty = 1$. In multivariate situation, for $\boldsymbol{k} = (k_1, k_2, ..., k_d)$ and $\boldsymbol{l} = (l_1, l_2, ..., l_d)$, we define

$$\mathcal{B}^d_{\boldsymbol{k},\boldsymbol{l}} = \prod_{i=1}^{d} B_r(k_i x_i - l_i). \tag{S.2}$$

as the multivariate cardinal B-splines, where r is suppressed for brevity. For the isotropic case $k \in \mathbb{N}$, we slightly abuse the notation and let $\mathcal{B}^d_{k,\boldsymbol{l}} = \prod_{i=1}^d B_r(kx_i - l_i)$ when all components of $\boldsymbol{k}$ equal to k. The following approximation result on splines for Hölder functions is well known.

Lemma S.1. *Suppose that $f \in \mathcal{C}^s(\Omega)$ and $\|f\|_{\mathcal{C}^s} \le B_4$. Let k be a positive integer, there are $\{a_{k,\boldsymbol{l}}\}$ with $j \in \mathbb{Z}_+$ and multi-index $\boldsymbol{l} \in \prod_{i=1}^d \{-r+1, -r+2, \ldots, k-1, k\}$, such that*

$$\left\| f - \sum_{\boldsymbol{l}} a_{k,\boldsymbol{l}} \mathcal{B}^d_{k,\boldsymbol{l}} \right\|_\infty \le c_{35} k^{-s} \|f\|_{\mathcal{C}^s},$$

where $a_{k,\boldsymbol{l}} \le c_{36}\|f\|_{\mathcal{C}^s}$ for any k and $\boldsymbol{l}$, c_{35} and c_{36} are constants no depending on k and f.

Note that the theory in this paper requires the functions in $\mathcal{F}_{nm}$ to be bounded. A simple way is to let $\mathcal{F}_{nm}$ be the truncated version of the functional space $\left\{\sum_{\boldsymbol{l}} \alpha_{k,\boldsymbol{l}} \mathcal{B}^d_{k,\boldsymbol{l}} : a_{k,\boldsymbol{l}} \in \mathbb{R}\right\}$ for suitable k, which brings no significant change to the VC dimension of $\mathcal{F}_{nm}$.

Another remedy is to set $\mathcal{F}_{nm}$ as $\left\{\sum_{\boldsymbol{l}} \alpha_{k,\boldsymbol{l}} \mathcal{B}^d_{k,\boldsymbol{l}} : a_{k,\boldsymbol{l}} \leq c_{36}\|f\|_{\mathcal{C}^s}\right\}$. However, the upper bound $c_{36}\|f\|_{\mathcal{C}^s}$ is usually unknown, which can be circumvented by relaxation to an increasing sequence, i.e.,

$$\mathcal{F}_{nm} = \left\{\sum_{\boldsymbol{l}} \alpha_{k,\boldsymbol{l}} \mathcal{B}^d_{k,\boldsymbol{l}} : a_{k,\boldsymbol{l}} \leq \log n\right\}. \tag{S.3}$$

Recall the definition of the regression spline estimator as in (2), while $\log n$ does not introduce much computational complexity to the optimization. Lemma S.1 implies the approximation error is of order $O(k^{-2s})$, and a direct application of Corollary 2.1 yields the following result.

Theorem S.7. *Consider the repeated measures model (1) and the estimator $\hat{f}_{nm}$ obtained by (2). Suppose that Assumptions 1, 2 and 3 hold, and set the candidate class as (S.3), then*

$$\mathcal{E}_{f^\circ}(\hat{f}_{nm}) \leq c_{37}(k^{-2s} + n^{-1}(\log n)^2 + (nm)^{-1}k^d(\log nm)^3),$$

for sufficiently large n, where c_{37} is a constant free of n, m and k.

Hence we choose $k \asymp O((nm)^{1/(2s+d)}(\log nm)^{-3/(2s+d)})$ for trade-off, leading to the prediction error $O(n^{-1}(\log n)^2 + (nm)^{-2s/(2s+d)}(\log nm)^{6s/(2s+d)})$ which matches the lower bound established in Theorem 3.2.

S.3 Additional Numerical Results

In this study, the Adam optimizer with a maximum iteration limit of 200 epochs is used for network training. We only employ the early stopping technique with patience 10 for regular-

ization. The network configurations are setting to be $(L, W) = (1, 50), (2, 100), (3, 200), (4, 400), (5, 600), (5, 800)$. Each example trains all networks with different depths and widths, then we select the model with the minimal validation error. Note that the proposed theory in this work assumes the order of LW. In the simulation study, LW ranges from 50 to 4000, which is in the typical selection range of tuning parameters.

For all simulations, the d-dimensional covariate X is generated from the uniform distribution on $\Omega \subseteq [0, 1]^d$. Recalling the model setting $Y_{ij} = f^\circ(X_{ij}) + U_i(X_{ij}) + \epsilon_{ij}$, we take the individual stochastic part $U_i(x) = \sum_{k=1}^{\infty} \sum_{l=1}^{d} \frac{\sqrt{3}\xi_{ikl}}{\sqrt{dk}} \cos(k\pi x_l)$ with independent scores $\xi_{ikl} \sim N(0, 0.1^2)$, and measurement errors $\epsilon_{ij} \sim N(0, 0.01^2)$. Now, we add three examples,

$$\text{Case S.1} : f^\circ(x) = x_1^2 x_2^3 + \log(1 + x_3) + \sqrt{1 + x_4 x_5} + \exp(x_5/2), \Omega = [0, 1]^5,$$

$$\text{Case S.2} : f^\circ(x) = (x_1^2 x_2^3 + \log(1 + x_3) + \sqrt{1 + x_4 x_5} + \exp(x_5/2))^2 - 12, \Omega = [0, 1]^5,$$

$$\text{Case S.3} : f^\circ(x) = \sum_{k_1, \ldots, k_5 \geq 1} a^{\max\{k_1, 2k_2, 2k_3, 2k_4, 2k_5\}} \prod_{l=1}^{5} \cos(2\pi l^{-1} b^{k_l} x_l), \Omega = [0, 1]^5.$$

Case S.1 and Case S.2 have been previously considered by Zhong et al. (2022) for the hierarchical model. Case S.3 serves as an anisotropic smooth counterpart of Case 1 because the coefficient decay rate in the last four components is higher. Results of the DNN estimator are shown in Figure S.4, and the comparisons with the regression spline method in Section S.2, RKHS method in Cai & Yuan (2011), and local linear method in Zhang & Wang (2016), are in Table S.4. Here, the RKHS method uses the comment Laplace kernel, and the local linear regression uses the comment Epanechnikov Kernel; the tuning parameters are selected by the validation set.

Similarly with Figure 1 in the paper, Figure S.4 also reflects the phase transition as n and m grow. It is observed that the DNN estimators learn these functions significantly better than Case 1 in the paper, due to their hierarchical structure and higher smoothness. Table S.4 then demonstrates the outstanding performance of the DNN estimators compared

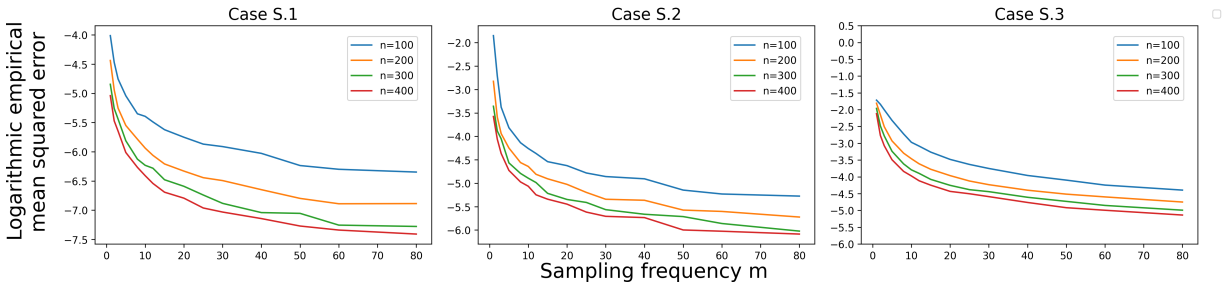


Figure S.4: Finite sample mean squared error of DNN estimators for $n = 100, 200, 300, 400$ and sampling frequencies $m = 1, 2, 3, 5, 8, 10, 12, 15, 20, 25, 30, 40, 50, 60, 80$. Each subplot portrays sampling frequencies (m) on the horizontal axis and logarithmic empirical mean squared error on the vertical axis.

Table S.4: Finite sample mean square error of the DNN estimator, naive regression spline estimator (for cases with $d \leq 5$), RKHS estimator and local linear regression estimator.

	Case S.1				Case S.2				Case S.3			
Sample size n	100		300		100		300		100		300	
Sampling frequency m	10	30	10	30	10	30	10	30	10	30	10	30
DNN	0.0045	0.0027	0.0020	0.0010	0.0141	0.0078	0.0075	0.0038	0.0513	0.0235	0.0226	0.0118
Spline	0.1143	0.0177	0.0182	0.0045	0.1108	0.0178	0.0183	0.0045	0.2603	0.0691	0.0690	0.0315
RKHS	0.0074	0.0039	0.0037	0.0019	0.0339	0.0159	0.0159	0.0079	0.0819	0.0493	0.0488	0.0280
Local Linear	0.7918	0.3556	0.3438	0.1099	1.1696	0.7856	0.6618	0.2783	0.1615	0.1320	0.1200	0.0941

to other methods. Finally, to further verify the rate more comprehensively, we add the verification for the influence of total observations nm and sample size n. In particular, Figure S.5 corresponds to Case S.1 with the sample size $n = 25, 100, 400$ and the overall observation size $nm = 400, 800, 1200, 2000, 3200, 4000, 4800, 6000, 8000, 12000, 16000, 20000, 24000, 32000$. The figure indicates that larger values of n lead to better estimation even when nm is the same, which provides more comprehensive empirical validation.

S.4 Technical Lemmas

This supplement contains the lemmas that are crucial for proof. Lemmas S.2-S.3 are crucial findings in the proof of Theorem 2.1 without involving cluster dependence. Lemmas S.4-S.6

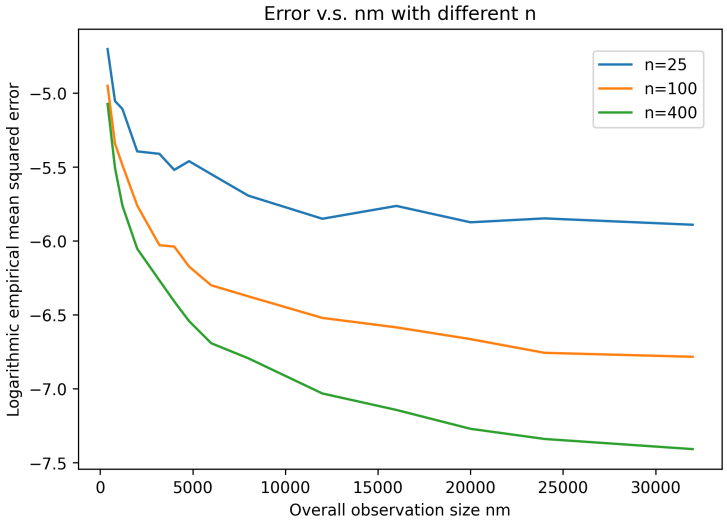


Figure S.5: Finite sample mean squared error of the DNN estimators for sample size $n = 25, 100, 400$ and overall observation size $nm = 400, 800, 1200, 2000, 3200, 4000, 4800, 6000, 8000, 12000, 16000, 20000, 24000, 32000$. Each subplot portrays the overall observation size (nm) on the horizontal axis and the logarithmic empirical mean squared error on the vertical axis.

play significant roles in the approximation theory of fully connected neural networks, while Lemmas S.7-S.10 serve as fundamental components for the empirical process. Additionally, Lemma S.11 is to prove the suboptimality of the linear estimator class for unknown hierarchical structures.

Lemma S.2. *Under the same model and assumptions as Theorem 2.1, we have*

$$\begin{aligned}&\left|\int_{\Omega}(\hat{f}-f^{\circ})^2 d\mathcal{P}_X - \frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\left(\hat{f}\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)^2\right| \\ \leq &\frac{1}{2}\int_{\Omega}(\hat{f}-f^{\circ})^2 d\mathcal{P}_X + c_{38}B_1^2 r^* + \frac{c_{39}B_1^2 t}{nm},\end{aligned}$$

with probability at least $1-\exp\{-t\}$*, where* c_{38} *and* c_{39} *are universal constants.*

Lemma S.3. *Under the same model and assumptions as Theorem 2.1, we have*

$$\begin{aligned}&\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\left(\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}-\mathbb{E}\left[\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}\right]\right)\left(\hat{f}\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)\right| \\ \leq &\frac{1}{16}\int_{\Omega}(\hat{f}-f^{\circ})^2 d\mathcal{P}_X + c_{40}\beta_\epsilon^2 r^* + \frac{c_{41}(B_1\beta_\epsilon+B_3^2)t}{nm}.\end{aligned}$$

with probability at least $1-\exp\{-t\}$*, where* $\mathcal{T}_{\beta_\epsilon}$ *is the truncation operator,* c_{40} *and* c_{41} *are universal constants.*

Lemma S.4. *Given positive integer* L, W *and* $K \leq L^2W^2$*, assume positive number* $\delta \leq \frac{1}{3K}$*. Then there is some ReLU neural network function* $f_{\mathcal{FNN}} \in \mathcal{FNN}(1, 4L+5, 4W+3)$ *that satisfies*

$$f_{\mathcal{FNN}}(x) = k, \quad \text{when } x \in \left[\frac{k}{K}, \frac{k+1}{K} - \delta \cdot I_{\{k \leq K-2\}}\right],$$

for $k = 0, 1, ..., K-1$*.*

Lemma S.5 (Proposition 4.4 in Lu et al. (2021))**.** *Given positive integer* L, W *and* $\lceil s \rceil$*. Suppose that* $y_l = 0$ *or* 1 *for* $l = 0, 1, 2, ..., L^2W^2 - 1$*. Then there exists a ReLU neural network function* $f_{\mathcal{FNN}} \in \mathcal{FNN}(1, 5(L+2)\log_2(4L), 16\lceil s \rceil (W+1)\log_2(8W))$ *that satisfies*

$$|f_{\mathcal{FNN}}(l) - y_l| \leq (LW)^{-2\lceil s \rceil},$$

for $l = 0, 1, 2, ..., L^2W^2 - 1$ *and* $0 \leq f_{\mathcal{FNN}}(x) \leq 1$ *for* $x \in \mathbb{R}$*.*

Lemma S.6. *Suppose that* $\mathcal{B}^d_{\boldsymbol{k},\boldsymbol{l}}$ *is defined by* (S.2)*. Then, for any* $L, W \geq 3$*, there exists a ReLU neural network function* $f_{\mathcal{FNN}} \in \mathcal{FNN}(d, c_{42}L, c_{43}W, 1)$ *satisfying*

$$|f_{\mathcal{FNN}}(x) - \mathcal{B}^d_{1,\mathbf{0}}| \leq c_{44}W^{-c_{45}L},$$

for every $x \in \{x : x_i \in [0, r/k_i], 1 \leq i \leq d\}$ *and* $c_{42}, c_{43}, c_{44}, c_{45}$ *are all constants independent of* L*,* W*.*

Compared with Lemma 1 in Suzuki (2018), this result has more flexible choices of the depth and width, since we use the approximation construction of multivariate product given by Lu et al. (2021).

The following lemmas are classic results of the empirical process, and their proofs will be omitted. For mathematical rigor, we define

$$\mathbb{E}\sup_{f\in\mathcal{F}} V(f) = \sup\left\{\mathbb{E}\sup_{f\in\mathcal{F}^*} V(f) : \mathcal{F}^* \subset \mathcal{F}, \mathcal{F}^* \text{ is finite}\right\}$$

for a random process V as in Talagrand (2014) when the index set $\mathcal{F}$ is uncountable.

Lemma S.7 (Comparison Theorem; see Theorem 4.12 in Ledoux & Talagrand (1991))**.** *Let $\{\Phi_i, 1 \leq i \leq n\}$ be functions that are b_1-Lipschitz and satisfy $\Phi_i(0) = 0$. Assume that $\{\sigma_i, 1 \leq i \leq n\}$ are Rademacher random variables not depending on others. Then for any bounded subset Ω' in $\mathbb{R}^n$, we have*

$$\mathbb{E}\left[\sup_{(t_1,\ldots,t_n)\in\Omega'}\left|\frac{1}{n}\sum_{i=1}^{n}\sigma_i\Phi_i\left(t_i\right)\right|\right] \leq 2b_1 \cdot \mathbb{E}\left[\sup_{(t_1,\ldots,t_n)\in\Omega'}\left|\frac{1}{n}\sum_{i=1}^{n}\sigma_i t_i\right|\right],$$

where the expectation is taken with respect to σ_i.

Lemma S.8 (Talagrand's inequality)**.** *Let $\{W_{i,f}, 1 \leq i \leq n, f \in \mathcal{F}\}$ be random variables independent cross i. Assume that for any $1 \leq i \leq n$ and $f \in \mathcal{F}$, we have $|W_{i,f}| \leq b_2$, $\mathbb{E}\left[W_{i,f}\right] = 0$ and $\mathbb{E}\left[W_{i,f}^2\right] \leq b_3$. Then with probability at least $1 - \exp\{-t\}$, the following inequality holds:*

$$\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}W_{i,f}\right| \leq 2\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}W_{i,f}\right|\right] + \sqrt{\frac{8b_3 t}{n}} + \frac{35b_2 t}{n}.$$

Lemma S.8 could be derived from Theorem 4 of Massart (2000). Unlike the usual versions of Talagrand's inequality with better constants, it does not impose the requirement that $W_i, 1 \leq i \leq n$ to be identically distributed.

Lemma S.9 (Dudley's entropy integral)**.** *Let $\mathcal{F}$ be a function class and suppose that $\sup_{f\in\mathcal{F}}\sum_{i=1}^{n}(f(X_i) - f^\circ(X_i))^2/n \leq b_4^2$ with given $\mathcal{X} = \{X_i, 1 \leq i \leq n\}$. Then we have*

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\sigma_i(f(X_i) - f^\circ(X_i))\right|\right] \leq \inf_{0\leq t\leq b_4}\left\{4t + \frac{12}{\sqrt{n}}\int_t^{b_4}\sqrt{\log\mathcal{N}\left(\varepsilon, \mathcal{F}|_{\mathcal{X}}, L^2\right)}d\varepsilon\right\},$$

where the expectation is taken with respect to σ_i.

Lemma S.10 (Corollary 2.2 in Bartlett et al. (2005))**.** *Let $\mathcal{F}$ be a function class and suppose that $\|f\|_\infty \leq 1$ for any $f \in \mathcal{F}$. Suppose that positive numbers r and t satisfy*

$$r \geq 10\mathbb{E}\left[\sup_{f\in\mathcal{F}(r)}\left|\frac{1}{n}\sum_{i=1}^{n}\sigma_i\left(f(X_i) - f^\circ(X_i)\right)\right|\right] + \frac{11t}{n},$$

where $\{X_i, 1 \le i \le n\}$ *are i.i.d. random variables. Then with probability at least* $1 - \exp\{-t\}$*, one holds:*

$$\mathcal{F}(r) \subseteq \left\{ f \in \mathcal{F} : \sum_{i=1}^{n} \left(f(X_i) - f^\circ(X_i)\right)^2 \le 2r \right\}.$$

Lemma S.11. *Consider the repeated measurement model* (1) *and assume that* $\mathcal{P}_X$ *has a uniform distribution on* $\Omega = [0,1]^d$*. Given a partition* $\mathcal{A}$ *of domain* Ω *with* $|\mathcal{A}| \asymp 2^K$ *where* $K \asymp \log(nm)$*, and each* $A \in \mathcal{A}$ *has the same measure* 2^{-K}*. For a constant* C*, define the event* $\mathcal{D}$ *by*

$$\max_{A \in \mathcal{A}} |\{X_{ij} \mid X_{ij} \in A, 1 \le i \le n, 1 \le j \le m\}| \le C 2^{-K} nm.$$

Consider the functional class $\mathcal{F}^\circ$ *defined on domain* Ω*. Assume that there are positive numbers* b_5, b_6 *and* b_7 *such that the following conditions holds.*

(i) $\forall A \in \mathcal{A}$*, there is a function* $g \in \mathcal{F}^\circ$ *satisfying* $g(x) \ge \frac{1}{2} b_5 b_6$ *for any* $x \in A$*.*

(ii) There is a $K' > 0$ *such that* $\sum_{i=1}^{n} \sum_{j=1}^{m} g(X_{ij})^2 \le nm b_7 b_5^2 2^{-K'}$ *for any* $g \in \mathcal{F}^\circ$ *on the event* $\mathcal{D}$*. Then we have*

$$\inf_{\tilde{f}_L} \sup_{\mathcal{P}_{Z,\epsilon} \in \mathcal{Q}_{Z,\epsilon}} \mathcal{E}_{f^\circ}(\tilde{f}_L) \gtrsim \frac{1}{n} + \min\left(\frac{b_6^2 2^{K'}}{b_7 nm} + b_5^2 b_6^2 2^{-K} \right),$$

where the infimum is taken over all linear estimators $\tilde{f}_L$ *based on the available data set* $\{(Y_{ij}, X_{ij}) : 1 \le i \le n, 1 \le j \le m\}$ *with form* $\tilde{f}_L(x) = \sum_{i=1}^{n} \sum_{j=1}^{m} Y_{ij} \varphi_{ij}(x, X_{11}, \ldots, X_{nm})$ *and the supremum is taken over all distributions* $\mathcal{P}_{Z,\epsilon}$ *satisfying the same assumptions in Theorem S.1.*

S.5 Proofs of Theorems and Corollaries

In this section, we provide proofs for the theorems and corollaries presented earlier in the paper. Due to space economy, we write $\int_\Omega (f - f^\circ)^2 d\mathcal{P}_X$ for the integral $\int_\Omega \left(f(x) - f^\circ(x)\right)^2 d\mathcal{P}_X$, $\mathcal{F} = \mathcal{F}_{nm}$, and abbreviate $\phi = \phi_{nm}$, $r^* = r^*_{nm}$, etc.

Proof of Theorem 2.1. The proof is divided into five steps. In the first step, we bound the prediction error $\mathcal{E}_{f^\circ}(\hat{f})$ by $\hat{\mathcal{E}}_{f^\circ}(\hat{f}) := \mathbb{E}[\sum_{i=1}^n \sum_{j=1}^m (\hat{f}(X_{ij}) - f^\circ(X_{ij}))^2/nm]$. According to Step 2, we bound $\hat{\mathcal{E}}_{f^\circ}(\hat{f})$ through several parts, which will be further analyzed in Step 3 and Step 4, respectively. Then the desired result follows from the conclusion in Step 5.

Step 1. Lemma S.2 implies that

$$\int_\Omega (\hat{f} - f^\circ)^2 d\mathcal{P}_X \leq \frac{2}{nm} \sum_{i=1}^n \sum_{j=1}^m \left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)^2 + 2c_{38} B_1^2 r^* + \frac{2c_{39} B_1^2 t}{nm},$$

holds with probability at least $1 - \exp\{-t\}$. By taking the expectation of the above result, we can further obtain

$$\mathcal{E}_{f^\circ}(\hat{f}) \leq \frac{2}{nm} \mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^m \left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)^2\right] + 2c_{38} B_1^2 r^* + \frac{2c_{39} B_1^2}{nm}. \tag{S.4}$$

Step 2. We now focus on the first term on the right side of (S.4). Recalling that $\hat{f}$ minimize the empirical loss, we have

$$\hat{\mathcal{E}}_{f^\circ}(\hat{f}) := \frac{1}{nm} \mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^m \left(Y_{ij} - \hat{f}(X_{ij})\right)^2\right] \leq \frac{1}{nm} \mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^m (Y_{ij} - f(X_{ij}))^2\right] \tag{S.5}$$

holds for any $f \in \mathcal{F}$. Substituting $Y_{ij} = f^\circ(X_{ij}) + U_i(X_{ij}) + \epsilon_{ij}$ into the above inequality, a bit of algebra shows

$$\hat{\mathcal{E}}_{f^\circ}(\hat{f}) \leq \int_\Omega (f - f^\circ)^2 \, d\mathcal{P}_X + \frac{2}{nm} \mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^m U_i(X_{ij}) \hat{f}(X_{ij})\right] + \frac{2}{nm} \mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^m \epsilon_{ij} \hat{f}(X_{ij})\right].$$

Because f is arbitrary and $\mathbb{E}\left[U_i(X_{ij}) f^\circ(X_{ij})\right] = \mathbb{E}\left[\epsilon_{ij} f^\circ(X_{ij})\right] = 0$, we have

$$\begin{aligned} \hat{\mathcal{E}}_{f^\circ}(\hat{f}) \leq \inf_{f \in \mathcal{F}} \int_\Omega (f - f^\circ)^2 \, d\mathcal{P}_X &+ \frac{2}{nm} \mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^m U_i(X_{ij}) \left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)\right] \\ &+ \frac{2}{nm} \mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^m \epsilon_{ij} \left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)\right]. \end{aligned} \tag{S.6}$$

Step 3. In this step, we mainly bound $\frac{1}{nm} \mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^m U_i(X_{ij}) \left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)\right]$ in (S.6). Recalling the truncation operator $\mathcal{T}_\beta$: $\mathcal{T}_\beta f(x) = f(x) I(|f(x)| \leq \beta) + \text{sign}(f) \beta I(|f(x)| >$

β), the triangle inequality implies

$$
\begin{aligned}
&\frac{1}{nm}\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{m}U_i(X_{ij})\left(\hat{f}(X_{ij})-f^{\circ}(X_{ij})\right)\right]\\
\leq&\mathbb{E}\left[\left|\frac{1}{n}\sum_{i=1}^{n}\left(\frac{1}{m}\sum_{j=1}^{m}\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(\hat{f}\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)-\int_{\Omega}\left(\mathcal{T}_{\beta_U}U_i\right)\left(\hat{f}-f^{\circ}\right)d\mathcal{P}_X\right)\right|\right]\\
&+\mathbb{E}\left[\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\left(U_i-\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(\hat{f}\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)-\int_{\Omega}\left(U_i-\mathcal{T}_{\beta_U}U_i\right)\left(\hat{f}-f^{\circ}\right)d\mathcal{P}_X\right|\right]\\
&+\mathbb{E}\left[\left|\frac{1}{n}\sum_{i=1}^{n}\int_{\Omega}U_i\left(\hat{f}-f^{\circ}\right)d\mathcal{P}_X\right|\right]\\
:=&\left(I\right)+\left(II\right)+\left(III\right).
\end{aligned}
\tag{S.7}
$$

Step 3.1. To control (I), we first study

$$
\sup_{f\in\mathcal{F}}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)-\int_{\Omega}\left(\mathcal{T}_{\beta_U}U_i\right)(f-f^{\circ})d\mathcal{P}_X}{\int_{\Omega}(f-f^{\circ})^2d\mathcal{P}_X+r}\right|.
$$

By the symmetrization result of the Rademacher process, we have

$$
\begin{aligned}
&\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)-\int_{\Omega}\left(\mathcal{T}_{\beta_U}U_i\right)(f-f^{\circ})d\mathcal{P}_X}{\int_{\Omega}(f-f^{\circ})^2d\mathcal{P}_X+r}\right|\right]\\
\leq&2\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\sigma_{ij}\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)}{\int_{\Omega}(f-f^{\circ})^2d\mathcal{P}_X+r}\right|\right],
\end{aligned}
$$

where σ_{ij}'s are i.i.d. Rademacher random variables. Then a peeling of the function class implies that it is less than

$$
\begin{aligned}
\leq&2\mathbb{E}\left[\sup_{f\in\mathcal{F}(r)}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\sigma_{ij}\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)}{\int_{\Omega}(f-f^{\circ})^2d\mathcal{P}_X+r}\right|\right]\\
&+\sum_{k=1}^{\infty}2\mathbb{E}\left[\sup_{f\in\mathcal{F}(4^kr)\setminus\mathcal{F}(4^{k-1}r)}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\sigma_{ij}\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)}{\int_{\Omega}(f-f^{\circ})^2d\mathcal{P}_X+r}\right|\right],
\end{aligned}
$$

which could be further bounded by

$$
\begin{aligned}
&\frac{2}{r}\mathbb{E}\left[\sup_{f\in\mathcal{F}(r)}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\sigma_{ij}\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)\right|\right]\\
&+\sum_{k=1}^{\infty}\frac{2}{(4^{k-1}+1)r}\mathbb{E}\left[\sup_{f\in\mathcal{F}(4^kr)}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\sigma_{ij}\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)\right|\right].
\end{aligned}
\tag{S.8}
$$

Taking the expectation conditional on $U_i, X_{ij}, 1 \le i \le n, 1 \le j \le m$ and utilizing Lemma S.7, we conclude that

$$\begin{aligned}
&\mathbb{E}\left[\sup_{f\in\mathcal{F}(r')}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\sigma_{ij}\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(f(X_{ij})-f^\circ(X_{ij})\right)\right|\right]\\
=&\mathbb{E}\left[\mathbb{E}\left[\sup_{f\in\mathcal{F}(r')}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\sigma_{ij}\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(f(X_{ij})-f^\circ(X_{ij})\right)\right|\middle| U, X\right]\right]\\
\le&2\beta_U\mathbb{E}\left[\sup_{f\in\mathcal{F}(r')}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\sigma_{ij}\left(f(X_{ij})-f^\circ(X_{ij})\right)\right|\right]\\
\le&2\beta_U\phi(r'),
\end{aligned}$$

for any $r' \ge r^*$. Combining the previous several inequalities, we thus obtain

$$\begin{aligned}
&\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(f-f^\circ\right)(X_{ij})-\int_\Omega\left(\mathcal{T}_{\beta_U}U_i\right)(f-f^\circ)d\mathcal{P}_X}{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X+r}\right|\right]\\
\le&\frac{4\beta_U\phi(r)}{r}+\sum_{k=1}^{\infty}\frac{4\beta_U\phi(4^kr)}{(4^{k-1}+1)r}\le\frac{4\beta_U\phi(r)}{r}+\sum_{k=1}^{\infty}\frac{2^{k+2}\beta_U\phi(r)}{(4^{k-1}+1)r}\le\frac{16\beta_U\phi(r)}{r}.
\end{aligned}\tag{S.9}$$

Furthermore, one can verify that

$$\mathbb{E}\left[\frac{\left(\mathcal{T}_{\beta_U}U_i\right)(X)\left(f(X)-f^\circ(X)\right)-\int_\Omega\left(\mathcal{T}_{\beta_U}U_i\right)(f-f^\circ)d\mathcal{P}_X}{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X+r}\right]=0,\tag{S.10}$$

$$\left|\frac{\left(\mathcal{T}_{\beta_U}U_i\right)(X)\left(f(X)-f^\circ(X)\right)-\int_\Omega\left(\mathcal{T}_{\beta_U}U_i\right)(f-f^\circ)d\mathcal{P}_X}{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X+r}\right|\le\frac{4B_1\beta_U}{r},\tag{S.11}$$

and

$$\mathbb{E}\left[\left|\frac{\left(\mathcal{T}_{\beta_U}U_i\right)(X)\left(f(X)-f^\circ(X)\right)-\int_\Omega\left(\mathcal{T}_{\beta_U}U_i\right)(f-f^\circ)d\mathcal{P}_X}{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X+r}\right|^2\right]\le\frac{\beta_U^2}{4r}.\tag{S.12}$$

By combining with (S.9), (S.10), (S.11) and (S.12) and checking the conditions required by Lemma S.8, we can conclude that, with probability at least $1-\exp\{-t\}$,

$$\begin{aligned}
&\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(\hat{f}(X_{ij})-f^\circ(X_{ij})\right)-\int_\Omega\left(\mathcal{T}_{\beta_U}U_i\right)\left(\hat{f}-f^\circ\right)d\mathcal{P}_X}{\int_\Omega(\hat{f}-f^\circ)^2d\mathcal{P}_X+r}\right|\\
\le&\sup_{f\in\mathcal{F}}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(f(X_{ij})-f^\circ(X_{ij})\right)-\int_\Omega\left(\mathcal{T}_{\beta_U}U_i\right)(f-f^\circ)d\mathcal{P}_X}{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X+r}\right|\\
\le&\frac{32\beta_U\phi(r)}{r}+\sqrt{\frac{2\beta_U^2t}{nmr}}+\frac{140B_1\beta_Ut}{nmr}.
\end{aligned}$$

We determine universal constants c_{46} and c_{47} such that $r_U(t) = c_{46}\beta_U^2 r^* + \frac{c_{47}(B_1+\beta_U)\beta_U t}{nm}$ satisfies

$$\frac{32\beta_U\phi(r_U)}{r_U} + \sqrt{\frac{2\beta_U^2 t}{nmr_U}} + \frac{140B_1\beta_U t}{nmr_U} \leq \frac{1}{32}.$$

Therefore, with probability at least $1-\exp\{-t\}$,

$$\begin{aligned}
&\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\left(\mathcal{T}_{\beta_U}U_i\right)(X_{ij})\left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right) - \int_\Omega \left(\mathcal{T}_{\beta_U}U_i\right)\left(\hat{f}-f^\circ\right)d\mathcal{P}_X\right| \\
\leq &\frac{1}{32}\int_\Omega (\hat{f}-f^\circ)^2 d\mathcal{P}_X + \frac{c_{46}\beta_U^2 r^*}{32} + \frac{c_{47}(B_1+\beta_U)\beta_U t}{32nm}.
\end{aligned}$$

Furthermore, we can bound the expectation

$$(I) \leq \frac{1}{32}\mathcal{E}_{f^\circ}(\hat{f}) + c_{48}\beta_U^2 r^* + \frac{c_{49}(B_1+\beta_U)\beta_U}{nm}, \tag{S.13}$$

where c_{48} and c_{49} are universal constants.

Step 3.2. In this substep, we focus on (II). Applying the absolute value inequality, we obtain

$$\begin{aligned}
(II) \leq &2\mathbb{E}\left[\left|\left(U_1 - \mathcal{T}_{\beta_U}U_1\right)(X_{11})\left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)\right|\right] \\
\leq &4B_1\mathbb{E}\left[|U_{11}|\, I(|U_{11}|>\beta_U)\right].
\end{aligned}$$

Applying the facts

$$|U_{11}| \leq 2B_2\exp\left\{\frac{|U_{11}|}{2B_2}\right\} \quad \text{and} \quad I(|U_{11}|>\beta_U) \leq \exp\left\{\frac{|U_{11}|-\beta_U}{2B_2}\right\}, \tag{S.14}$$

we have

$$(II) \leq 8B_1B_2\mathbb{E}\left[\exp\left\{\frac{|U_{11}|}{B_2} - \frac{\beta_U}{2B_2}\right\}\right] = 8B_1B_2\exp\left\{-\frac{\beta_U}{2B_2}\right\}. \tag{S.15}$$

Step 3.3. Now we consider (III). Using the GM-QM inequality,

$$\begin{aligned}
(III) = &\mathbb{E}\left[\left|\int_\Omega\left(\frac{1}{n}\sum_{i=1}^{n}U_i\right)\left(\hat{f}-f^\circ\right)d\mathcal{P}_X\right|\right] \\
\leq &8\mathbb{E}\left[\int_\Omega\left(\frac{1}{n}\sum_{i=1}^{n}U_i\right)^2 d\mathcal{P}_X\right] + \frac{1}{32}\mathbb{E}\left[\int_\Omega\left(\hat{f}-f^\circ\right)^2\mathcal{P}_X\right].
\end{aligned}$$

Because $\mathbb{E}\left[U(X)^2\right] \leq \mathbb{E}\left[B_2^2 \exp\left\{U(X)/B_2\right\}\right] \leq B_2^2$, we have $\mathbb{E}\left[\int_\Omega \left(\sum_{i=1}^n U_i/n\right)^2 d\mathcal{P}_X\right] \leq B_2^2/n$. Substituting this into the earlier inequality yields

$$(III) \leq \frac{8B_2^2}{n} + \frac{1}{32}\mathcal{E}_{f^\circ}(\hat{f}). \tag{S.16}$$

Substituting (S.13), (S.15) and (S.16) into (S.7) and setting $\beta_U = 2B_2 \log n$, we have

$$\begin{aligned} &\frac{1}{nm}\mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^m U_i(X_{ij})\left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)\right] \\ \leq &\frac{1}{16}\mathcal{E}_{f^\circ}(\hat{f}) + \frac{c_{50}(B_1^2 + B_2^2)}{n} + \frac{5c_{49}(B_1^2 + B_2^2 \log^2 n)}{nm} + 4c_{48}B_2^2 r^* \log^2 n. \end{aligned} \tag{S.17}$$

Step 4. In this step, we aim to control $\frac{1}{nm}\mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^m \epsilon_{ij}\left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)\right]$. The proof in this step is similar to Step 3. The main gap is that ϵ_{ix} is not integrable with respect to x. Thus it cannot be given by a direct extension of Step 3. Using the triangle inequality, we have

$$\begin{aligned} &\frac{1}{nm}\mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^m \epsilon_{ij}\left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)\right] \\ \leq &\mathbb{E}\left[\left|\frac{1}{n}\sum_{i=1}^n \left(\frac{1}{m}\sum_{j=1}^m \left(\mathcal{T}_{\beta_\epsilon}\epsilon_{ij} - \mathbb{E}\left[\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}\right]\right)\left(\hat{f}\left(X_{ij}\right) - f^\circ\left(X_{ij}\right)\right)\right)\right|\right] \\ &+ \left|\mathbb{E}\left[\frac{1}{nm}\sum_{i=1}^n \sum_{j=1}^m \left(\epsilon_{ij} - \mathcal{T}_{\beta_\epsilon}\epsilon_{ij} + \mathbb{E}\left[\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}\right]\right)\left(\hat{f}\left(X_{ij}\right) - f^\circ\left(X_{ij}\right)\right)\right]\right| \\ := &(IV) + (V). \end{aligned} \tag{S.18}$$

Step 4.1. Lemma S.3 implies that, with probability at least $1 - \exp\{-t\}$,

$$\begin{aligned} &\left|\frac{1}{nm}\sum_{i=1}^n \sum_{j=1}^m \left(\mathcal{T}_{\beta_\epsilon}\epsilon_{ij} - \mathbb{E}\left[\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}\right]\right)\left(\hat{f}\left(X_{ij}\right) - f^\circ\left(X_{ij}\right)\right)\right| \\ \leq &\frac{1}{16}\int_\Omega (\hat{f} - f^\circ)^2 d\mathcal{P}_X + c_{40}\beta_\epsilon^2 r^* + \frac{c_{41}(B_1\beta_\epsilon + B_3^2)t}{nm}. \end{aligned}$$

Further, by taking the expectation of the previous result, we have

$$(IV) \leq \frac{1}{16}\mathcal{E}_{f^\circ}(\hat{f}) + c_{40}\beta_\epsilon^2 r^* + \frac{c_{41}(B_1\beta_\epsilon + B_3^2)}{nm}. \tag{S.19}$$

Step 4.2. Next we bound (V). Note that $\mathbb{E}[\epsilon_{ij}] = 0$ yields $|\mathbb{E}[\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}]| = |\mathbb{E}[\epsilon_{ij} I(|\epsilon_{ij}| > \beta_\epsilon)]|$. Then

$$\begin{aligned}(V) \leq & 2B_1\mathbb{E}[|\epsilon_{ij} - \mathcal{T}_{\beta_\epsilon}\epsilon_{ij} + \mathbb{E}[\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}]|] \leq 4B_1\mathbb{E}[|\epsilon_{11}| I(|\epsilon_{11}| > \beta_\epsilon)] \\ \leq & 4B_1\mathbb{E}\left[2B_3 \exp\left\{\frac{|\epsilon_{11}|}{2B_3}\right\} \exp\left\{\frac{|\epsilon_{11}| - \beta_\epsilon}{2B_3}\right\}\right] \leq 8B_1B_3 \exp\left\{\frac{-\beta_\epsilon}{2B_3}\right\}.\end{aligned} \tag{S.20}$$

Substituting (S.19) and (S.20) into (S.18) and setting $\beta_\epsilon = 2B_3 \log nm$, we have

$$\begin{aligned}& \frac{1}{nm}\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{m}\epsilon_{ij}\left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)\right] \\ \leq & \frac{1}{16}\mathcal{E}_{f^\circ}(\hat{f}) + \frac{c_{51}(B_1^2 + B_3^2\log^2 nm)}{nm} + c_{52}B_3^2 r^* \log^2 nm,\end{aligned} \tag{S.21}$$

where c_{51} and c_{52} are universal constants.

Step 5. Plugging (S.17)(S.21) into (S.6) yields

$$\begin{aligned}\hat{\mathcal{E}}_{f^\circ}(\hat{f}) \leq & \inf_{f\in\mathcal{F}}\int_\Omega (f - f^\circ)^2 d\mathcal{P}_X + \frac{1}{4}\mathcal{E}_{f^\circ}(\hat{f}) + \frac{c_{53}(B_1^2 + B_2^2)}{n} \\ & + \frac{c_{54}(B_1^2 + B_2^2\log n + B_3^2\log nm)}{nm} + c_{55}r^*(B_2^2\log n + B_3^2\log nm)).\end{aligned} \tag{S.22}$$

Substituting (S.22) into (S.4) and noting that all constants c_k used in this proof are universal, the desired result follows. Thus we complete the proof. □

Remark S.5.1. *It is known that, in practice, the exact empirical risk minimizer may not be attainable due to optimization error, while the non-convex function class could also trap the gradient-descent-based optimization methods into local minima. Thus it is meaningful to study the theoretical performance of estimators when they cannot strictly reach the global minimum. Like in Schmidt-Hieber (2020), we define the optimization error of an estimator $\hat{f}_{nm}$ by*

$$\Delta_{nm}(\hat{f}_{nm}) := \mathbb{E}\left[\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\left(Y_{ij} - \hat{f}_{nm}(X_{ij})\right)^2 - \inf_{f\in\mathcal{F}_{nm}}\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{m}(Y_{ij} - f(X_{ij}))^2\right],$$

which measures the difference between the estimator $\hat{f}_{nm}$ and the global empirical risk minimum over the chosen class $\mathcal{F}_{nm}$. It is easy to show that $\Delta_{nm}(\hat{f}_{nm}) \geq 0$ and $\Delta_{nm}(\hat{f}_{nm}) = 0$

if and only if $\hat{f}_{nm}$ *satisfies* (2). *Consider the optimization error* $\Delta_{nm}(\hat{f}_{nm})$ *defined above and under the same model and assumptions as Theorem 2.1, there are universal positive constants* c_{56}, c_{57} *and* c_{58} *such that*

$$
\begin{aligned}
& c_{56}\Delta_{nm}(\hat{f}_{nm}) - c_{57}\left(\frac{B_1^2+B_2^2}{n} + (r_{nm}^* + \frac{1}{nm})\iota_{nm}\right) \\
& \leq \mathcal{E}_{f^\circ}(\hat{f}_{nm}) \\
& \leq c_{58}\left(\inf_{f\in\mathcal{F}_{nm}} \int_\Omega (f-f^\circ)^2 \, d\mathcal{P}_X + \Delta_{nm}(\hat{f}_{nm}) + \frac{B_1^2+B_2^2}{n} + (r_{nm}^* + \frac{1}{nm})\iota_{nm}\right),
\end{aligned}
$$

where $\iota_{nm} = B_1^2 + B_2^2(\log n)^2 + B_3^2(\log nm)^2$.

We first prove the upper bound. It requires a reconstruction of Step 2 in the proof of Theorem 2.1. Obviously,

$$
\inf_{f\in\mathcal{F}_{nm}} \frac{1}{n}\sum_{i=1}^n\sum_{j=1}^m (Y_{ij} - f(X_{ij}))^2 \leq \frac{1}{nm}\sum_{i=1}^n\sum_{j=1}^m (Y_{ij} - f(X_{ij}))^2
$$

holds for any $f \in \mathcal{F}$. *Therefore we have*

$$
\frac{1}{nm}\mathbb{E}\left[\sum_{i=1}^n\sum_{j=1}^m \left(Y_{ij} - \hat{f}(X_{ij})\right)^2\right] \leq \frac{1}{nm}\mathbb{E}\left[\sum_{i=1}^n\sum_{j=1}^m (Y_{ij} - f(X_{ij}))^2\right] + \Delta_{nm}(\hat{f}_{nm}).
$$

Because f *is arbitrary and* $\mathbb{E}\left[U_i(X_{ij})\hat{f}^\circ(X_{ij})\right] = \mathbb{E}\left[\epsilon_{ij}\hat{f}^\circ(X_{ij})\right] = 0$, *we have*

$$
\begin{aligned}
& \frac{1}{nm}\mathbb{E}\left[\sum_{i=1}^n\sum_{j=1}^m \left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)^2\right] \\
& \leq \inf_{f\in\mathcal{F}} \int_\Omega (f-f^\circ)^2 \, d\mathcal{P}_X + \frac{2}{nm}\mathbb{E}\left[\sum_{i=1}^n\sum_{j=1}^m U_i(X_{ij})\left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)\right] \\
& \quad + \frac{2}{nm}\mathbb{E}\left[\sum_{i=1}^n\sum_{j=1}^m \epsilon_{ij}\left(\hat{f}(X_{ij}) - f^\circ(X_{ij})\right)\right] + \Delta_{nm}(\hat{f}_{nm}).
\end{aligned}
$$

The rest is the same as the proof of Theorem 2.1.

For the lower bound. Let

$$
\tilde{f}_{nm} \in \arg\min_{f\in\mathcal{F}_{nm}} \frac{1}{nm}\sum_{i=1}^n\sum_{j=1}^m (Y_{ij} - f(X_{ij}))^2
$$

be a exact empirical risk minimizer. A few algebra shows

$$\begin{aligned}
&\frac{1}{nm}\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{m}\left(\hat{f}(X_{ij})-f^{\circ}(X_{ij})\right)^2\right]-\frac{1}{nm}\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{m}\left(\tilde{f}(X_{ij})-f^{\circ}(X_{ij})\right)^2\right]\\
=&\Delta_{nm}(\hat{f}_{nm})+\frac{2}{nm}\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{m}(U_i(X_{ij})+\epsilon_{ij})(\hat{f}(X_{ij})-f^{\circ}(X_{ij}))\right]\\
&-\frac{2}{nm}\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{m}(U_i(X_{ij})+\epsilon_{ij})(\hat{f}(X_{ij})-f^{\circ}(X_{ij}))\right].
\end{aligned}$$

Using the results of Step 3 and Step 4 in the Proof of Theorem 2.1, we have

$$\begin{aligned}
&\frac{1}{nm}\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{m}\left(\hat{f}(X_{ij})-f^{\circ}(X_{ij})\right)^2\right]-\frac{1}{nm}\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{m}\left(\tilde{f}(X_{ij})-f^{\circ}(X_{ij})\right)^2\right]\\
\geq&\Delta_{nm}(\hat{f}_{nm})-\frac{1}{4}\left(\mathbb{E}\left[\int_{\Omega}(\hat{f}-f^{\circ})^2d\mathcal{P}_X\right]+\mathbb{E}\left[\int_{\Omega}(\tilde{f}-f^{\circ})^2d\mathcal{P}_X\right]\right) \qquad . \quad \text{(S.23)}\\
&-c_{59}\left(\frac{B_1^2+B_2}{n}+(r_{nm}^{*}+\frac{1}{nm})\iota_{nm}\right)
\end{aligned}$$

Similar to the proof of Lemma S.2 and Step 1 in the proof of Theorem 2.1, we can derive that

$$\mathbb{E}\left[\int_{\Omega}(\tilde{f}-f^{\circ})^2d\mathcal{P}_X\right]\leq\frac{2}{nm}\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{m}\left(\tilde{f}(X_{ij})-f^{\circ}(X_{ij})\right)^2\right]+2c_{38}B_1^2r^{*}+\frac{2c_{39}B_1^2}{nm}.$$

and

$$\mathbb{E}\left[\int_{\Omega}(\tilde{f}-f^{\circ})^2d\mathcal{P}_X\right]\geq\frac{2}{3nm}\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{m}\left(\tilde{f}(X_{ij})-f^{\circ}(X_{ij})\right)^2\right]-\frac{2c_{38}B_1^2r^{*}}{3}-\frac{2c_{39}B_1^2}{3nm}.$$

Substituting the above two formulas and their counterparts about $\hat{f}$ together into the (S.23). Noting that the term $\mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{m}\left(\tilde{f}(X_{ij})-f^{\circ}(X_{ij})\right)^2\right]$ is nonnegative, the desired lower bound follows.

Proof of Corollary 2.1 . (i) Let $\mathcal{G}=\{g=f/B_1 : f\in\mathcal{F}\}$, then every $g\in\mathcal{G}$ satisfies $\|g\|_{\infty}\leq 1$. Denote $g^{\circ}=f^{\circ}/B_1$. Then the estimation problem becomes

$$\hat{g}_{nm}=\arg\min_{g\in\mathcal{G}_{nm}}\sum_{i=1}^{n}\sum_{j=1}^{m}\left(Y_{ij}/B_1-g(X_{ij})\right)^2.$$

Denote $B_2' = B_2/B_1$ and $B_3' = B_3/B_1$. By Theorem 2.1, we have

$$\begin{aligned}\mathcal{E}_{g^\circ}(\hat{g}_{nm}) \leq & c_1 \left(\inf_{g\in\mathcal{G}} \int_\Omega (g-g^\circ)^2 \, d\mathcal{P}_X + \frac{1+B_2'^2}{n} \right. \\ & \left. + \left(r_{nm}^*(\mathcal{G}) + \frac{1}{nm} \right) (1 + B_2'^2 \log n + B_3'^2 \log nm) \right).\end{aligned} \tag{S.24}$$

Let

$$r_1 = \sup \left\{ r : 10\mathbb{E}\left[\sup_{g\in\mathcal{G}(r)} \left| \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m \sigma_{ij} \left(g(X_{ij}) - g^\circ(X_{ij}) \right) \right| \right] + \frac{11 \log nm}{nm} \geq r \right\}.$$

Because $\mathbb{E}\left[\sup_{g\in\mathcal{G}(r)} \left| \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m \sigma_{ij} \left(g(X_{ij}) - g^\circ(X_{ij}) \right) \right| \right]$ is monotonic non-decreasing with respect to r, simple analysis shows that r_1 satisfies

$$r_1 = 10\mathbb{E}\left[\sup_{g\in\mathcal{G}(r_1)} \left| \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m \sigma_{ij} \left(g(X_{ij}) - g^\circ(X_{ij}) \right) \right| \right] + \frac{11 \log nm}{nm}. \tag{S.25}$$

(In fact, we can deduce a contradiction if replacing the equal sign in the above equation with either the greater or less.) For any $r \geq r_1$, consider

$$\begin{aligned}\mathcal{R}_{nm}\left(\mathcal{G}(r)\right) = & \mathbb{E}\left[\sup_{g\in\mathcal{G}(r)} \left| \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m \sigma_{ij} \left(g(X_{ij}) - g^\circ(X_{ij}) \right) \right| \right] \\ \leq & \mathbb{E}\left[\sup_{g\in\hat{\mathcal{G}}(2r)} \left| \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m \sigma_{ij} \left(g(X_{ij}) - g^\circ(X_{ij}) \right) \right| \right] \\ & + \mathbb{E}\left[\sup_{g\in\mathcal{G}(r)\setminus\hat{\mathcal{G}}(2r)} \left| \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m \sigma_{ij} \left(g(X_{ij}) - g^\circ(X_{ij}) \right) \right| \right].\end{aligned}$$

where $\hat{\mathcal{G}}(2r) = \left\{ g \in \mathcal{G} : \sum_{i=1}^n \sum_{j=1}^m \left(\hat{g}(X_{ij}) - g^\circ(X_{ij}) \right)^2 / nm \leq 2r \right\}$ is a random set depending on samples. Applying Lemma S.9, we have

$$\begin{aligned}\sup_{g\in\hat{\mathcal{G}}(2r)} \left| \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m \sigma_{ij} \left(g(X_{ij}) - g^\circ(X_{ij}) \right) \right| \leq & \frac{4}{nm} + \frac{12}{\sqrt{nm}} \int_{1/nm}^{\sqrt{2r}} \sqrt{\log \mathcal{N}\left(\varepsilon, \mathcal{G}|_{\mathcal{X}}, L^2\right)} d\varepsilon \\ \leq & \frac{4}{nm} + \frac{12\sqrt{2r}}{\sqrt{nm}} \sqrt{\log \mathcal{N}\left(1/nm, \mathcal{G}|_{\mathcal{X}}, L^2\right)}.\end{aligned}$$

Hence, the first term is bounded by

$$\frac{4}{nm} + \frac{12\sqrt{2r}}{\sqrt{nm}} \mathbb{E}\left[\sqrt{\log \mathcal{N}\left(1/nm, \mathcal{G}|_{\mathcal{X}}, L^2\right)} \right].$$

Also, the second term could be bounded by

$$\begin{aligned}
&\mathbb{E}\left[\sup_{g\in\mathcal{G}(r)\setminus\hat{\mathcal{G}}(2r)}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\sigma_{ij}\left(g(X_{ij})-g^{\circ}(X_{ij})\right)\right|\right]\\
\leq&\mathbb{P}\left(\left\{\exists g\in\mathcal{G}(r):\ \frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\sigma_{ij}\left(g(X_{ij})-g^{\circ}(X_{ij})\right)>2r\right\}\right)\cdot 2\\
\leq&\frac{2}{nm},
\end{aligned}$$

where the last inequality is given by Lemma S.10 and (S.25). Hence,

$$\mathcal{R}_{nm}\left(\mathcal{G}(r)\right)\leq\frac{6}{nm}+\frac{12\sqrt{2r}}{\sqrt{nm}}\mathbb{E}\left[\sqrt{\mathcal{N}\left(1/nm,\mathcal{G}|_{\mathcal{X}},L^2\right)}\right]$$

is bounded by a sub-root function for $r\geq r_1$. In addition, recalling (S.25), we have

$$\begin{aligned}
r_1=&10\mathbb{E}\left[\sup_{g\in\mathcal{G}(r_1)}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\sigma_{ij}\left(g(X_{ij})-g^{\circ}(X_{ij})\right)\right|\right]+\frac{11\log nm}{nm}\\
\leq&\frac{60}{nm}+\frac{120\sqrt{2r_1}}{\sqrt{nm}}\mathbb{E}\left[\sqrt{\log\mathcal{N}\left(1/nm,\mathcal{G}|_{\mathcal{X}},L^2\right)}\right]+\frac{11\log nm}{nm}\\
\leq&\frac{60}{nm}+\frac{r_1}{2}+\frac{120^2}{nm}\mathbb{E}\left[\log\mathcal{N}\left(1/nm,\mathcal{G}|_{\mathcal{X}},L^2\right)\right]+\frac{11\log nm}{nm}.
\end{aligned}$$

That implies, there exists a constant c_{60}

$$r_1\leq\frac{c_{60}}{nm}\left(\mathbb{E}\left[\log\mathcal{N}\left(1/nm,\mathcal{G}|_{\mathcal{X}},L^2\right)\right]+\log nm\right).$$

Substituting $r^*_{nm}(\mathcal{G})\leq r_1$ to (S.24), we have

$$\begin{aligned}
\mathcal{E}_{g^{\circ}}(\hat{g}_{nm})\leq c_{61}\Bigg(&\inf_{g\in\mathcal{G}}\int_{\Omega}\left(g-g^{\circ}\right)^2d\mathcal{P}_X+\frac{1+B_2'^2}{n}\\
&+\frac{1}{nm}\left(\mathbb{E}\left[\log\mathcal{N}\left(1/nm,\mathcal{G}|_{\mathcal{X}},L^2\right)\right]+\log nm\right)\left(1+B_2'^2\log n+B_3'^2\log nm\right)\Bigg).
\end{aligned}$$

Recalling that $B_1\mathcal{G}=\mathcal{F}$ yields $\mathcal{N}\left(1/nm,\mathcal{G}|_{\mathcal{X}},L^2\right)=\mathcal{N}\left(B_1/nm,\mathcal{F}|_{\mathcal{X}},L^2\right)$, the desired result follows.

(ii) Applying Theorem 9.4 in Györfi et al. (2002) and the relationship of covering number and packing number (see, for example, Lemma 9.2 in Györfi et al. (2002)), we have

$$\log\mathcal{N}\left(\frac{B_1}{nm},\mathcal{F}_{nm}|_{\mathcal{X}},L^2\right)\leq\mathrm{VCdim}(\mathcal{F}_{nm})\log(2e(nm)^2\log 3e(nm)^2)+\log 3.$$

Recalling that $\log nm\geq 1$, then (i) yields the desired result. □

Proof of Corollary 2.2 . Theorem 6 in Bartlett et al. (2019) tells us that the VC dimension of a ReLU neural with length L and width W can be bounded by

$$\text{VCdim} \leq c_{62} L^2 W^2 \log(LW),$$

where c_{62} is a constant. Combining this conclusion and (ii) of Corollary 2.1 implies the desired result immediately.

□

Proof of Theorem 3.1. By Corollary 3.1 of Jiao et al. (2023), we know there is a neural network $f'_{\mathcal{FNN}}$ with width $c_{63} L \log L$ and depth $c_{64} W \log W$ that can uniformly approximate f° with an accuracy of $c_{65} (LW)^{-2s/d}$, where the constants do not depend on L and W. Using a truncation that could be realized by ReLU units easily, we can substitute the approximation error into Corollary 2.2. Specifying L, W satisfying

$$LW = \lfloor c_7 (nm)^{d/(4s+2d)} (\log nm)^{-4d/(2s+d)} \rfloor$$

and the desired result follows. □

Proof of Theorem 3.2. Given the sample sizes n and m, we define

$$J = \lfloor \log_2(nm)/(2s+d) \rfloor$$

and c_{66} to be a constant determined later. To consider B-splines with disjoint support sets, we introduce a multi-index set

$$\mathcal{V} = \left\{ \boldsymbol{v} = (v_1, v_2, ..., v_d) : -r+1 \leq v_i \leq 2^J \text{ and } v_i = rt+1 \text{ for some } t \in \mathbb{Z} \right\}. \tag{S.26}$$

One can verify that, for $\boldsymbol{v}, \boldsymbol{v}' \in \mathcal{V}, \boldsymbol{v} \neq \boldsymbol{v}'$, the support sets of $\mathcal{B}^d_{2^J, \boldsymbol{v}}$ and $\mathcal{B}^d_{2^J, \boldsymbol{v}'}$ are disjoint. Then we consider the following set

$$\left\{ \sum_{\boldsymbol{v}} a_{J,\boldsymbol{v}} 2^{-Js} c_{66} \mathcal{B}^d_{2^J, \boldsymbol{v}} : a_{J,\boldsymbol{v}} \in \{0,1\}, \boldsymbol{v} \in \mathcal{V} \right\}. \tag{S.27}$$

It can be shown that the cardinality of $\mathcal{V}$ is given by $|\mathcal{V}| = (\lceil 2^J/r \rceil + 1)^d \geq 2^{Jd}/r^d$. Moreover, for any $\boldsymbol{a} = (a_{J,\boldsymbol{v}} : v \in \mathcal{V})$, the function $f_{\boldsymbol{a}} = \sum_{\boldsymbol{v}} a_{J,\boldsymbol{v}} 2^{-Js} c_{66} \mathcal{B}^d_{2^J,\boldsymbol{v}}$ in the set defined by (S.27) belongs to $\mathcal{C}^s$, since it is a linear combination of B-splines with order r and disjoint support sets. In fact,

$$\|f_{\boldsymbol{a}}\|_{\mathcal{C}^s} \leq c_{67} c_{66} 2^{-Js} \|\mathcal{B}^d_{2^J,\boldsymbol{0}}\|_{\mathcal{C}^s} \leq c_{68} c_{66} < \infty.$$

Note that $f_{\boldsymbol{a}} - f_{\boldsymbol{a}'} = \sum_{\boldsymbol{v}} (a_{J,\boldsymbol{v}} - a'_{J,\boldsymbol{v}}) 2^{-Js} c_{66} \mathcal{B}^d_{2^J,\boldsymbol{v}}$ for two different functions $f_{\boldsymbol{a}}$ and $f_{\boldsymbol{a}'}$ from the set (S.27). By the construction of $\mathcal{V}$, we have

$$\|f_{\boldsymbol{a}} - f_{\boldsymbol{a}'}\|^2_{L^2(\Omega)} = 2^{-2Js-Jd} c_{69} c_{66}^2 \, \mathrm{Ham}(\boldsymbol{a}, \boldsymbol{a}'),$$

where Ham is the standard Hamming distance and $c_{69} = \|\mathcal{B}^d_{1,\boldsymbol{0}}\|^2_{L^2(\Omega)}$. Using the Varshamov-Gilbert lemma, there is a subset $\mathcal{A} \subseteq \{0,1\}^{|\mathcal{V}|}$ of cardinality $|\mathcal{A}| \geq \exp\{|\mathcal{V}|/8\}$, such that, for any $\boldsymbol{a}, \boldsymbol{a}' \in \mathcal{A}, \boldsymbol{a} \neq \boldsymbol{a}'$, $\mathrm{Ham}(\boldsymbol{a}, \boldsymbol{a}') \geq |\mathcal{V}|/8$ holds.

Note that proving the lower bound only requires consideration of a specific case. Thus, we can first assume Z is a Gaussian process and ϵ is a Gaussian noise with variance B_3. Then, the joint distribution of $(Y_{11}, Y_{12}, ..., Y_{1m}, ..., Y_{n1}, Y_{n2}, ..., Y_{nm})$ is multivariate normal distribution with mean

$$\mu_{X,f_{\boldsymbol{a}}} = (f_{\boldsymbol{a}}(X_{11}), f_{\boldsymbol{a}}(X_{11}), ..., f_{\boldsymbol{a}}(X_{1m}), ..., f_{\boldsymbol{a}}(X_{n1}), f_{\boldsymbol{a}}(X_{n1}), ..., f_{\boldsymbol{a}}(X_{nm})) \tag{S.28}$$

and covariance matrix

$$C_X = \mathrm{diag}\left\{(\mathrm{Cov}(Z_1(X_{1j}, X_{1k})))_{1 \leq j,k \leq m}\right\}_{1 \leq i \leq n} + B_3 I, \tag{S.29}$$

conditional on $f^\circ = f_{\boldsymbol{a}}$ and measurement points $\{X_{ij}, 1 \leq j \leq m\}$. Hence, we can bound

the Kullback-Leibler divergence of two probability measures from $\mathcal{P}_{f_{\boldsymbol{a}}}$ to $\mathcal{P}_{f_{\boldsymbol{a}'}}$ by

$$\begin{aligned}
\mathrm{KL}(\mathcal{P}_{f_{\boldsymbol{a}}}, \mathcal{P}_{f_{\boldsymbol{a}'}}) =& \mathbb{E}\left[(\mu_{X,f_{\boldsymbol{a}}} - \mu_{X,f_{\boldsymbol{a}'}}) C_X^{-1} (\mu_{X,f_{\boldsymbol{a}}} - \mu_{X,f_{\boldsymbol{a}'}})\right] \\
\leq& \mathbb{E}\left[(\mu_{X,f_{\boldsymbol{a}}} - \mu_{X,f_{\boldsymbol{a}'}}) (B_3 I)^{-1} (\mu_{X,f_{\boldsymbol{a}}} - \mu_{X,f_{\boldsymbol{a}'}})\right] \\
=& nm B_3^{-1} \left\| f_{\boldsymbol{a}} - f_{\boldsymbol{a}'} \right\|_{L^2(\Omega)}^2 \\
=& 2^{-2Js - Jd} c_{70} c_{66}^2 nm B_3^{-1} \operatorname{Ham}(\boldsymbol{a}, \boldsymbol{a}') \\
\leq& c_{71} c_{66}^2 B_3^{-1} (nm)^{d/(2s+d)},
\end{aligned}$$

where c_{71} is a constant independent of n and m. Also,

$$\begin{aligned}
\left\| f_{\boldsymbol{a}} - f_{\boldsymbol{a}'} \right\|_{L^2(\Omega)}^2 =& 2^{-2Js - Jd} c_{69} c_{66}^2 \operatorname{Ham}(\boldsymbol{a}, \boldsymbol{a}') \\
\geq& c_{72} c_{66}^2 (nm)^{-2s/(2s+d)},
\end{aligned}$$

where c_{72} is also a constant independent of n and m. Applying Fano's inequality yields

$$\begin{aligned}
\max_{a \in \mathcal{A}} \mathbb{E}_{f_{\boldsymbol{a}}}\Big[\int_\Omega (\tilde{f} - f_{\boldsymbol{a}})^2 d\mathcal{P}_X \Big] &= \max_{a \in \mathcal{A}} \mathbb{E}_{f_{\boldsymbol{a}}} \left[\left\| \tilde{f} - f_{\boldsymbol{a}} \right\|_{L^2(\Omega)}^2 \right] \\
&\geq \frac{c_{72} c_{66}^2 (nm)^{-2s/(2s+d)}}{4} \left(1 - \frac{c_{71} c_{66}^2 B_3^{-1} (nm)^{d/(2s+d)} + \log 2}{2^{-d-3} r^{-d} (nm)^{d/(2s+d)}} \right).
\end{aligned}$$

Now we specify the value of c_{66} such that $c_{68} c_{66} \leq 1$ and $c_{72} c_{66}^2 \leq 2^{-d-5} r^{-d}$. Then we have

$$1 - \frac{c_{71} c_{66}^2 B_3^{-1} (nm)^{d/(2s+d)} + \log 2}{2^{-d-3} r^{-d} (nm)^{d/(2s+d)}} \geq \frac{1}{2},$$

for nm large enough. Taking into account the finite cases where nm is small, we know that there is a constant c_{73} independent of n and m such that

$$\max_{a \in \mathcal{A}} \mathbb{E}_{f_{\boldsymbol{a}}} \left[\int_\Omega (\tilde{f} - f_{\boldsymbol{a}})^2 d\mathcal{P}_X \right] \geq c_{73} (nm)^{-2s/(2s+d)}.$$

for any estimator $\tilde{f}$. This implies

$$\inf_{\tilde{f}} \sup_{\mathcal{P}_{Z,\epsilon} \in \mathcal{Q}_{Z,\epsilon}} \mathbb{E}_{f^\circ} \left[\int_\Omega (\tilde{f} - f^\circ)^2 d\mathcal{P}_X \right] \geq c_{73} (nm)^{-2s/(2s+d)}. \tag{S.30}$$

To show the remaining $1/n$ term in the lower bound, it now suffices to consider Z_i as constant functions over $x \in \Omega$ and the measurement noise ϵ_{ij} as zero. Assume that

$\mathrm{Var}(Z_i(X)|X) = B_2$. Then the problem becomes the mean estimation from n i.i.d. observations. It is well known that the optimal rate is B_2/n. It implies

$$\inf_{\tilde{f}} \sup_{\mathcal{P}_{Z,\epsilon} \in \mathcal{Q}_{Z,\epsilon}} \mathbb{E}_{f^\circ} \left[\int_\Omega (\tilde{f} - f^\circ)^2 d\mathcal{P}_X \right] \geq B_2 n^{-1}. \tag{S.31}$$

Then combining (S.30) and (S.31) completes the proof. □

Proof of Theorem 4.1. The proof is shown by induction on l.

First, the case $l = 1$ is given by Corollary 3.1 in Jiao et al. (2023). Assume that the result holds for cases $< l$. Now we consider the case for l. Suppose that $f \in \mathcal{H}^{l,\mathcal{G}}$ with $\mathcal{G} = ((s, K); \mathcal{G}_1, ..., \mathcal{G}_K)$. Then we have $\gamma_i = \gamma(\mathcal{G}_i) \leq \gamma / \min(1, s)$ and $s/K \leq \gamma$. By the definition of hierarchical composition model, we know $f(x)$ is equal to some $g(h_1(x), \ldots, h_K(x))$, where $h_i \in \mathcal{H}^{l-1,\mathcal{G}_i}(\Omega)$ and $g \in \mathcal{C}^s$. Since $h_i \in \mathcal{H}^{l-1,\gamma_i}(\Omega)$, there exist positive numbers M_l such that $h_i(x) \leq M_l$ for each $x \in \Omega$ and $i : 1 \leq i \leq K$. Note that

$$g(h_1(x), \ldots, h_K(x)) = g\left(2M_l \cdot \frac{h_1(x) + M_l}{2M_l} - \frac{1}{2}, \ldots, 2M_l \cdot \frac{h_K(x) + M_l}{2M_l} - \frac{1}{2} \right)$$

and functions $g(2M_l \cdot * - 1/2, \ldots, 2M_l \cdot * - 1/2) \in \mathcal{C}^s([0,1]^K)$, $h_i/M_l \in \mathcal{H}^{l-1,\gamma_i}(\Omega)$, without loss of generality, we assume that $0 \leq h_i(x) \leq 1$ for each i and $x \in \Omega$.

Given any $L, W \geq 2$. By the inductive hypothesis, there exist neural networks

$$\check{h}_i \in \mathcal{FNN}(d, c_{74} L \log L, c_{75} W \log W, 1),$$

and a constant c_{76} such that

$$|\check{h}_i(x) - h_i(x)| \leq c_{76} (LW)^{-2\gamma_i}, \tag{S.32}$$

for every $x \in \Omega$. Also, according to Corollary 3.1 in Jiao et al. (2023), there exist a $\check{g}_i \in \mathcal{FNN}(d, c_{77} L \log L, c_{78} W \log W, B_1)$ and a constant c_{79} such that

$$|\check{g}(x) - g(x)| \leq c_{79} (LW)^{-2s/k}, \tag{S.33}$$

for each $x \in [0,1]^K$. Then

$$\begin{aligned}
&\left|\check{g}\left(\check{h}_1(x),\ldots,\check{h}_K(x)\right) - g\left(h_1(x),\ldots,h_K(x)\right)\right| \\
\leq &\left|\check{g}\left(\check{h}_1(x),\ldots,\check{h}_K(x)\right) - g\left(\check{h}_1(x),\ldots,\check{h}_K(x)\right)\right| \\
&+ \left|g\left(\check{h}_1(x),\ldots,\check{h}_K(x)\right) - g\left(h_1(x),\ldots,h_K(x)\right)\right|.
\end{aligned}$$

By (S.33), the first term is bounded by $c_{79}\left(LW\right)^{-2s/K} \leq c_{79}\left(LW\right)^{-2\gamma}$. Additionally, by recalling (S.32) and $g \in \mathcal{C}^s$, the second term is bounded by

$$c_{80}(\sum_{i=1}^{K}\left(LW\right)^{-2\gamma_i})^{\min(s,1)} \leq c_{81}\left(LW\right)^{-2\min_i \gamma_i \min(s,1)} \leq c_{82}\left(LW\right)^{-2\gamma}.$$

To obtain the final result, we only need to check the size of $\check{g} \circ \left(\check{h}_1,\ldots,\check{h}_K\right)$. The width is bounded by $\max\left(c_{75}W\log W, c_{78}W\log W\right) = c_{83}W\log W$ and the depth is bounded by $c_{74}L\log L + c_{77}L\log L = c_{84}L\log L$. This completes the proof.

□

Proof of Theorem 4.2. Combining Theorem S.3 and Corollary 2.2 yields the desired result immediately. □

Proof of Theorem 4.3. By the assumption, we know that there is a node $G_1^\iota = (s_0, d_0)$ in the tree $\mathcal{G}$ such that $s_0^*/d_0 = \gamma$ and $d_0 \leq d$. Assume that $G_1^{\iota-1}, G_2^{\iota-1}, ..., G_{d_0}^{\iota-1}$ are children nodes of G_1^ι in the tree $\mathcal{G}$. Let $\mathcal{G}_1^\iota$ and $\mathcal{G}_1^{\iota-1}, \mathcal{G}_2^{\iota-1}, ..., \mathcal{G}_{d_0}^{\iota-1}$ be the maximal sub-trees whose roots are G_1^ι and $G_1^{\iota-1}, G_2^{\iota-1}, ..., G_{d_0}^{\iota-1}$, respectively. Let $h_k(x) = x_k$ be the k-th component of x. It is easy to verify that $h_k \in \mathcal{H}^{\iota-1,\mathcal{G}_k}(\Omega)$ for $1 \leq k \leq d_0$. Therefore,

$$g\left(h_1(x), h_2(x), ..., h_{d_0}(x)\right) \in \mathcal{H}^{\iota,\mathcal{G}_1^\iota}(\Omega)$$

for any $g : [0,1]^{d_0} \to \mathbb{R}$ satisfying $\|g\|_{\mathcal{C}^{s_0}} \leq M$. Denote $\varrho = \prod_{(s,K)\in\mathcal{A}(G_1^\iota)} \min(1,s)$ where $\mathcal{A}(G_1^\iota)$ denotes the ancestor nodes of G_1^ι. Iteratively applying the fact that $m(x) = (h_k(x))^{\min(1,s)} \in \mathcal{C}^s$ yields

$$f = \left(g\left(h_1(x), h_2(x), ..., h_K(x)\right)\right)^\varrho \in \mathcal{H}^{l,\mathcal{G}}(\Omega). \tag{S.34}$$

Given the sample sizes n and m, let $J = \lfloor \log_2(nm)/(2s_0^* + d_0) \rfloor$ and c_{85} be a constant to specific later. Let

$$\mathcal{V} = \left\{ \boldsymbol{v} = (v_1, v_2, ..., v_{d_0}) : -r + 1 \leq v_i \leq 2^J \text{ and } v_i = rt + 1 \text{ for some } t \in \mathbb{Z} \right\}.$$

We consider the following set

$$\left\{ f \text{ is of the form (S.34) with } g = \sum_{\boldsymbol{v}} a_{J,\boldsymbol{v}} 2^{-Js_0} c_{85} \mathcal{B}^{d_0}_{2^J,\boldsymbol{v}}, a_{J,\boldsymbol{v}} \in \{0,1\}, \boldsymbol{v} \in \mathcal{V} \right\}.$$

Note that the support sets of $\mathcal{B}^{d_0}_{2^J,\boldsymbol{v}}$ and $\mathcal{B}^{d_0}_{2^J,\boldsymbol{v}'}$ are disjoint for $\boldsymbol{v}, \boldsymbol{v}' \in \mathcal{V}, \boldsymbol{v} \neq \boldsymbol{v}'$, hence $\|g_a\|_{\mathcal{C}^{s_0}} \leq 2\|2^{-Js_0} c_{85} \mathcal{B}^{d_0}_{2^J,\boldsymbol{v}}\|_{\mathcal{C}^{s_0}} \leq c_{86} c_{85}$. Moreover, we could rewrite the above set as

$$\left\{ f = \sum_{\boldsymbol{v}} a_{J,\boldsymbol{v}} 2^{-Js^*} c_{85}^{\varrho} (\mathcal{B}^{d_0}_{2^J,\boldsymbol{v}})^{\varrho} : a_{J,\boldsymbol{v}} \in \{0,1\}, \boldsymbol{v} \in \mathcal{V} \right\}.$$

The remaining discussion to obtain the $(nm)^{-\gamma/(2\gamma+1)}$ term in the lower bound is entirely similar to Theorem 3.2. Additionally, the proof of n^{-1} term is also the same. Finally, by combining the results, we complete the proof. □

Proof of Theorem S.2. Define the high order difference operator $\Delta^r_{\boldsymbol{h}}(f)$ by

$$\Delta^r_{\boldsymbol{h}}(f)(x) = \sum_{i=0}^{r} \binom{r}{i} (-1)^{r-i} f(x + i\boldsymbol{h}),$$

when $x + ih$ are all within the domain. Otherwise, define $\Delta^r_{\boldsymbol{h}}(f)(x) = 0$ if some $x + ih$ are out of the domain. For $p \in (0, \infty]$, we define the r-th modulus of smoothness of f by

$$\omega_{r,p}(f, \boldsymbol{t}) = \sup_{|h_j| \leq t_j} \|\Delta^r_{\boldsymbol{h}}(f)\|_p,$$

when $\boldsymbol{t} = (t_1, t_2, ..., t_d)$ is a positive component vector. We also define the i-th partial modulus of smoothness by

$$\omega^{(i)}_{r,p}(f, t) = \omega_{r,p}(f, t\boldsymbol{e}_i),$$

when t is a positive real number.

Step 1. We first approximate the anisotropic $\boldsymbol{s}$-Hölder function by B-spline quasi-interpolant representation. By the definition of modulus and (4.76) in Giné & Nickl (2015),

$$\begin{aligned}\sup_{t>0}\frac{\omega_{r,\infty}^{(i)}(f,t)}{t^{s_i}} \leq & 2^{r-\lfloor s_i\rfloor-1}\sup_{t>0}\frac{\omega_{\lfloor s_i\rfloor+1,\infty}^{(i)}(f,t)}{t^{s_i}}\\ \leq & 2^{r-\lfloor s_i\rfloor-1}\sup_{t>0}\frac{\omega_{1,\infty}^{(i)}(\partial_i^{\lfloor s_i\rfloor}f,t)}{t^{s_i-\lfloor s_i\rfloor}}\\ = & 2^{r-\lfloor s_i\rfloor-1}\sup_{\substack{x,x'\in\Sigma\\ x'=x+te_i,t\neq 0}}\frac{\left|\partial_i^{\lfloor s_i\rfloor}f(x)-\partial_i^{\lfloor s_i\rfloor}f(x')\right|}{\|x-x'\|^{s_i-\lfloor s_i\rfloor}}.\end{aligned}$$

Therefore, there exists c_{87} not depending on f such that

$$\|f\|_\infty+\sum_{i=1}^d\sup_{t>0}\frac{\omega_{r,\infty}^{(i)}(f,t)}{t^{s_i}}\leq c_{87}\|f\|_{\mathcal{C}^{\boldsymbol{s}}}<\infty. \tag{S.35}$$

Denote $s_{\max}=\max_{1\leq j\leq d}s_j$, $s_{\min}=\min_{1\leq j\leq d}s_j$, $s_i'=s_{\min}/s_i$ and the harmonic mean

$$\tilde{s}=\frac{d}{\frac{1}{s_1}+\frac{1}{s_2}+...+\frac{1}{s_d}}.$$

Let

$$\boldsymbol{k}(j)=\left(2^{\lfloor js_1'\rfloor},2^{\lfloor js_2'\rfloor},...,2^{\lfloor js_d'\rfloor}\right),$$

for each positive integer j. By Theorem 3.2.6 in Leisner (2003), there are real numbers $\{a_{j,\boldsymbol{l}}\}$ for $j\in\mathbb{Z}_+$ and multi-index $\boldsymbol{l}\in\prod_{i=1}^d\left\{-r+1,-r+2,\ldots,2^{\lfloor js_i'\rfloor}-1,2^{\lfloor js_i'\rfloor}\right\}$, such that

$$\left\|f-\sum_{\boldsymbol{l}}a_{j,\boldsymbol{l}}\mathcal{B}^d_{\boldsymbol{k}(j),\boldsymbol{l}}\right\|_\infty\leq c_{88}\omega_{r,\infty}\left(f,(2^{-js_1'},2^{-js_2'},...,2^{-js_d'})\right).$$

By the definition of $\omega_{r,\infty}$ and $\omega_{r,\infty}^{(i)}$, we know

$$\omega_{r,\infty}\left(f,(2^{-js_1'},2^{-js_2'},...,2^{-js_d'})\right)\leq\sum_{i=1}^d\omega_{r,\infty}^{(i)}(f,2^{-js_i'}).$$

Further, using (S.35), we have

$$\sum_{i=1}^d\omega_{r,\infty}^{(i)}(f,2^{-js_i'})\leq\sum_{i=1}^d c_{89}2^{-js_i's_i}\|f\|_{\mathcal{C}^{\boldsymbol{s}}}=c_{90}2^{-js_{\min}}\|f\|_{\mathcal{C}^{\boldsymbol{s}}}.$$

Combining several inequalities derived above yields that there is a constant c_{91} independent of f and j such that

$$\left\| f - \sum_{\boldsymbol{l}} a_{j,\boldsymbol{l}} \mathcal{B}^d_{\boldsymbol{k}(j),\boldsymbol{l}} \right\|_\infty \leq c_{91} 2^{-j s_{\min}} \|f\|_{\mathcal{C}^{\boldsymbol{s}}} \tag{S.36}$$

holds. In addition, by Lemma 3.2.2 and Lemma 3.2.5 in Leisner (2003), it follows that one exists c_{92} and c_{93} such that

$$|a_{j,\boldsymbol{l}}| \leq c_{92} \left\| \sum_{\boldsymbol{l}} a_{j,\boldsymbol{l}} \mathcal{B}^d_{\boldsymbol{k}(j),\boldsymbol{l}} \right\|_\infty \leq c_{93} \|f\|_{\mathcal{C}^{\boldsymbol{s}}} \tag{S.37}$$

for each j and $\boldsymbol{l}$.

Step 2. Given L and W, let $J = \lfloor \frac{2\tilde{s}}{d s_{\min}} \log_2(LW) \rfloor$. We will approximate $\sum_{\boldsymbol{l}} a_{J,\boldsymbol{l}} \mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{l}}$ by a fully connected ReLU neural network with the desired size. Since J has been determined, let $\delta \in \left(0, \min_{1\leq i\leq d} 2^{-\lfloor J s_i' \rfloor}/3\right]$ be a small number to be determined latter. Define a trifling region $\Upsilon(J,\delta) \subseteq \Omega$ by

$$\Upsilon(J,\delta) := \bigcup_{t=1}^d \left\{ x = (x_1, x_2, ..., x_d) : x \in \Omega \text{ and } x_t \in \cup_{k=1}^{2^{J s_t'}-1} \left(\frac{k}{2^{J s_t'}} - \delta, \frac{k}{2^{J s_t'}} \right) \right\}.$$

Note that

$$2^{J s_i'} \leq (LW)^{\frac{2\tilde{s}}{d s_{\min}} s_i'} = (LW)^{\frac{2\tilde{s}}{d s_i}} \leq (LW)^2,$$

where the last inequality is due to $1/s_i \leq \sum_{i=1}^d 1/s_i = d/\tilde{s}$. Hence, applying Lemma S.4, we get $\check{f}_{1,i} \in \mathcal{FNN}(1, 4L+5, 4W+3)$ satisfying

$$\check{f}_{1,i}(x) = \kappa_i, \quad \text{when } x \in \left[\frac{\kappa_i}{2^{J s_i'}}, \frac{\kappa_i + 1}{2^{J s_i'}} - \delta \cdot I_{\{\kappa_i \leq 2^{J s_i'} - 2\}} \right],$$

for $\kappa_i = 0, 1, ..., 2^{J s_i'} - 1$ and $i = 1, 2, ..., d$. Let

$$\check{f}_2(x) = (\check{f}_{1,1}(x), \check{f}_{1,2}(x), ..., \check{f}_{1,d}(x)).$$

Intuitively, this function divides $\Omega \backslash \Upsilon(J,\delta)$ into $\prod_{i=1}^d 2^{\lfloor J s_i' \rfloor}$ blocks, i.e. for any $x = (x_1, x_2, ..., x_d) \in \Omega \backslash \Upsilon(J,\delta)$, $\check{f}_2(x) = (\kappa_1, \kappa_2, ..., \kappa_d)$ if and only if

$$x \in \prod_{i=1}^d \left[\frac{\kappa_i}{2^{J s_i'}}, \frac{\kappa_i + 1}{2^{J s_i'}} - \delta \cdot I_{\{\kappa_i \leq 2^{J s_i'} - 2\}} \right].$$

Note that the domain of a B-spline is bounded. Let

$$\mathcal{U} := \left\{ \boldsymbol{u} = (u_1, u_2, ...u_d) \in \mathbb{Z}^d : 0 \le u_i \le r-1 \text{ for } 1 \le i \le d \right\}.$$

For any $x \in \Omega \backslash \Upsilon(J, \delta)$, assume that $\check{f}_2(x) = (\kappa_1, \kappa_2, ..., \kappa_d) := \boldsymbol{\kappa}$. Then we know that $\mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{l}}(x) \neq 0$ if and only if $\boldsymbol{\kappa} - \boldsymbol{l} \in \mathcal{U}$. We should approximate the coefficients of these terms for each x. To apply Lemma S.5, we should first construct a function $\check{f}_3(\boldsymbol{\kappa}) = \sum_{i=1}^d \kappa_i 2^{J(i-1)}$ that maps all possible values of $\boldsymbol{\kappa}$ onto a subset of $\mathbb{Z}$ one by one. Because $\boldsymbol{\kappa}$ only takes $\prod_{i=1}^d 2^{\lfloor J s_i' \rfloor} \le \prod_{i=1}^d (LW)^{\frac{2\tilde{s}}{ds_i}} = (LW)^2$ different values, by Lemma S.5, there are $\check{f}_{4,\boldsymbol{u}} \in \mathcal{FNN}\left(1, 5(L+2)\log_2(4L), 16\lceil s_{\max} \rceil (W+1)\log_2(8W)\right)$ satisfying

$$\left| \check{f}_{4,\boldsymbol{u}} \circ \check{f}_3(\boldsymbol{\kappa}) - a_{j,\boldsymbol{\kappa}+\boldsymbol{u}} \right| \le c_{94}(LW)^{-2\lceil s_{\max} \rceil}, \tag{S.38}$$

for $\boldsymbol{u} \in \mathcal{U}$.

By Lemma S.6, there is a $\check{f}_5 \in \mathcal{FNN}\left(d, c_{42}L, c_{43}W, 1\right)$ satisfying $|\check{f}_5(x) - \mathcal{B}^d_{1,\boldsymbol{0}}| \le c_{44}W^{-c_{45}L}$, for every $x \in \{x : x_i \in [0, r/k_i], 1 \le i \le d\}$. Also, by Lemma 4.2 in Lu et al. (2021), there is a function $\check{f}_6$ implemented by a ReLU FNN with depth L and width $9W+1$ could approximate product function in a determined bounded domain and suitable precision. Let

$$\check{f}_7(x) = \sum_{\boldsymbol{u} \in \mathcal{U}} \check{f}_6\left(\check{f}_{4,\boldsymbol{u}} \circ \check{f}_3 \circ \check{f}_2(x), \check{f}_5(2^{\boldsymbol{k}(J)}x - \check{f}_2(x) - \boldsymbol{u}) \right).$$

The approximation error could be bounded by

$$\begin{aligned}
&\left| \check{f}_7(x) - \sum_{\boldsymbol{l}} a_{J,\boldsymbol{l}} \mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{l}} \right| = \left| \check{f}_7(x) - \sum_{\boldsymbol{u} \in \mathcal{U}} a_{J,\boldsymbol{\kappa}+\boldsymbol{u}} \mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{\kappa}+\boldsymbol{u}} \right| \\
\le &\sum_{\boldsymbol{u} \in \mathcal{U}} \left| \check{f}_6\left(\check{f}_{4,\boldsymbol{u}} \circ \check{f}_3 \circ \check{f}_2(x), \check{f}_5(2^{\boldsymbol{k}(J)}x - \check{f}_2(x) - \boldsymbol{u}) \right) - a_{J,\boldsymbol{\kappa}+\boldsymbol{u}} \mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{\kappa}+\boldsymbol{u}} \right| \\
\le &\sum_{\boldsymbol{u} \in \mathcal{U}} \left| \check{f}_6\left(\check{f}_{4,\boldsymbol{u}} \circ \check{f}_3 \circ \check{f}_2(x), \check{f}_5(2^{\boldsymbol{k}(J)}x - \check{f}_2(x) - \boldsymbol{u}) \right) - \check{f}_{4,\boldsymbol{u}} \circ \check{f}_3 \circ \check{f}_2(x) \times \check{f}_5(2^{\boldsymbol{k}(J)}x - \check{f}_2(x) - \boldsymbol{u}) \right| \\
&+ \sum_{\boldsymbol{u} \in \mathcal{U}} \left| \check{f}_{4,\boldsymbol{u}} \circ \check{f}_3 \circ \check{f}_2(x) \times \check{f}_5(2^{\boldsymbol{k}(J)}x - \check{f}_2(x) - \boldsymbol{u}) - a_{J,\boldsymbol{\kappa}+\boldsymbol{u}} \mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{\kappa}+\boldsymbol{u}} \right|.
\end{aligned}$$

By (S.37), (94) and Lemma 4.2 in Lu et al. (2021), the first term can be bounded by $c_{95}W^{-L}$. The second term is bounded by

$$\begin{aligned}
&\sum_{\boldsymbol{u}\in\mathcal{U}} \left|\check{f}_{4,\boldsymbol{u}} \circ \check{f}_3 \circ \check{f}_2(x) \times \check{f}_5(2^{\boldsymbol{k}(J)}x - \check{f}_2(x) - \boldsymbol{u}) - a_{J,\boldsymbol{\kappa}+\boldsymbol{u}}\mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{\kappa}+\boldsymbol{u}}\right| \\
\leq &\sum_{\boldsymbol{u}\in\mathcal{U}} \left|\check{f}_{4,\boldsymbol{u}} \circ \check{f}_3 \circ \check{f}_2(x) - a_{J,\boldsymbol{\kappa}+\boldsymbol{u}}\right| \cdot \left|\check{f}_5(2^{\boldsymbol{k}(J)}x - \check{f}_2(x) - \boldsymbol{u})\right| \\
&+ \sum_{\boldsymbol{u}\in\mathcal{U}} \left|\check{f}_5(2^{\boldsymbol{k}(J)}x - \check{f}_2(x) - \boldsymbol{u}) - \mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{\kappa}+\boldsymbol{u}}\right| \cdot \left|a_{J,\boldsymbol{\kappa}+\boldsymbol{u}}\right| \\
\leq &c_{96}(LW)^{-2\lceil s_{\max}\rceil} + c_{97}W^{-c_{45}L}.
\end{aligned}$$

Therefore,

$$\left|\check{f}_7(x) - \sum_{\boldsymbol{l}} a_{J,\boldsymbol{l}}\mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{l}}\right| \leq c_{95}W^{-L} + c_{96}(LW)^{-2\lceil s_{\max}\rceil} + c_{97}W^{-c_{45}L} \leq c_{98}(LW)^{-2\tilde{s}/d},$$

for $x \in \Omega\backslash\Upsilon(J,\delta)$. Recalling (S.36) and $J = \lfloor \frac{2\tilde{s}}{ds_{\min}} \log_2(LW) \rfloor$, we have

$$\left|f(x) - \sum_{\boldsymbol{l}} a_{J,\boldsymbol{l}}\mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{l}}(x)\right| \leq c_{91}2^{-Js_{\min}}\|f\|_{\mathcal{C}^{\boldsymbol{s}}} \leq c_{99}(LW)^{-2\tilde{s}/d}.$$

Combining the previous two results by triangle inequality, it follows that

$$\left|\check{f}_7(x) - f(x)\right| \leq c_{100}(LW)^{-2\tilde{s}/d},$$

where $x \in \Omega\backslash\Upsilon(J,\delta)$ and c_{100} is a constant independent of L and W. And the size of an implementation can be determined by Figure S.6 easily.

Step 3. The median function of three arguments could be implemented by a ReLU neural network with width 14 and depth 2 because

$$\begin{aligned}
\operatorname{mid}(t_1,t_2,t_3) = &\operatorname{ReLU}(t_1+t_2+t_3) - \operatorname{ReLU}(-t_1-t_2-t_3) \\
&- \max\{t_1,t_2,t_3\} - \min\{t_1,t_2,t_3\},
\end{aligned}$$

where

$$\begin{aligned}
\max\{t_1,t_2,t_3\} &= \max\{\max\{t_1,t_2\},t_3\} \\
&= \max\{\max\{t_1,t_2\}, \operatorname{ReLU}(t_3) - \operatorname{ReLU}(-t_3)\},
\end{aligned}$$

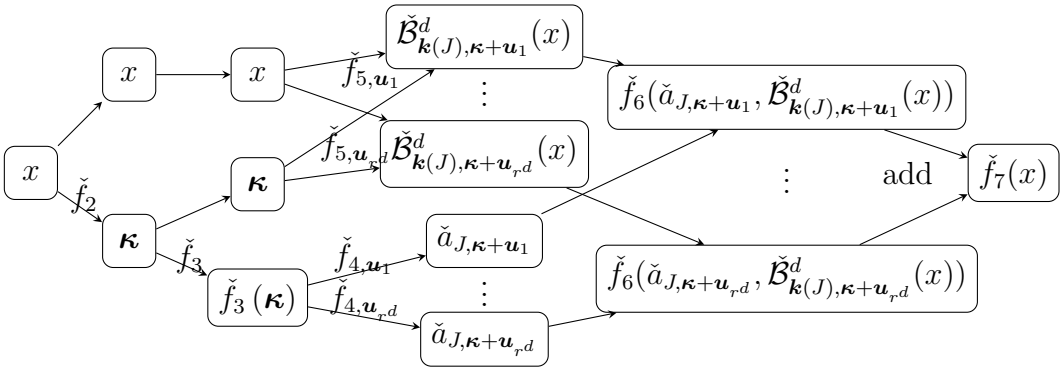


Figure S.6: An illustration of the sub-network architecture implementing $\check{f}_7(x)$. Here, we denote $\check{f}_5(2^{\boldsymbol{k}(J)}x - \check{f}_2(x) - \boldsymbol{u})$, $\check{f}_5(2^{\boldsymbol{k}(J)}x - \check{f}_2(x) - \boldsymbol{u})$ and $\check{f}_{4,\boldsymbol{u}} \circ \check{f}_3(\boldsymbol{\kappa})$ as $\check{f}_{5,\boldsymbol{u}}(x)$, $\check{\mathcal{B}}^d_{\boldsymbol{k}(J),\boldsymbol{\kappa}+\boldsymbol{u}}(x)$ and $\check{a}_{J,\boldsymbol{\kappa}+\boldsymbol{u}}$, respectively.

$$\max\{t_1, t_2\} = \frac{1}{2}(\,\mathrm{ReLU}\,(t_1 + t_2) - \mathrm{ReLU}\,(-t_1 - t_2) + \mathrm{ReLU}\,(t_1 - t_2) + \mathrm{ReLU}\,(t_2 - t_1)).$$

and the min function can be implemented similarly.

Set $\check{f}_{8,0} = \check{f}_7$, then we inductively define

$$\check{f}_{8,i}(x) := \mathrm{mid}\left(\check{f}_{8,i-1}\left(x - \delta e_i\right), \check{f}_{8,i-1}(x), \check{f}_{8,i-1}\left(x + \delta e_i\right)\right),$$

for $i = 1, 2, ..., d$. We also introduce

$$\Upsilon_i(J,\delta) := \bigcup_{t=i+1}^{d} \left\{ x = (x_1, x_2, ..., x_d) : x \in \Omega \text{ and } x_t \in \cup_{k=1}^{2^{Js'_t}-1} \left(\frac{k}{2^{Js'_t}} - \delta, \frac{k}{2^{Js'_t}} \right) \right\},$$

for $i = 0, 1, ..., d$. It is trivial that $\Upsilon_0(J,\delta) = \Upsilon(J,\delta)$ and $\Upsilon_d(J,\delta) = \emptyset$.

To control the approximation error $\check{f}_{8,d}$ in Σ, we prove the following stronger claim by induction

$$|\check{f}_{8,i}(x) - f(x)| \le c_{100}(LW)^{-2\tilde{s}/d} + i\|f\|_{\mathcal{C}^{\boldsymbol{s}}}\delta^{\min(s_{\min},1)} \tag{S.39}$$

for $x \in \Omega \backslash \Upsilon_i(J,\delta)$ and $i = 0, 1, ..., d$. In fact, when $i = 0$, it is given by the construction in the previous step. Suppose that the claim holds for $< i$, consider the case for $i = t$. Recalling the definition of $\Upsilon_i(J,\delta)$, we know there are at most one of $x - \delta e_i$, x and $x + \delta e_i$

is in $\Upsilon_i(J,\delta)$. Hence, by inductive assumption, there is at most one of $\check{f}_{8,i-1}(x-\delta e_i)-f(x-\delta e_i)$, $\check{f}_{8,i-1}(x)-f(x)$ and $\check{f}_{8,i-1}(x+\delta e_i)-f(x+\delta e_i)$ is larger than $c_{100}(LW)^{-2\tilde{s}/d}+(i-1)\|f\|_{\mathcal{C}^s}\delta^{\min(s_{\min},1)}$. Combining with $|f(x\pm\delta e_i)-f(x)|\leq\|f\|_{\mathcal{C}^s}\delta^{\min(s_i,1)}\leq\|f\|_{\mathcal{C}^s}\delta^{\min(s_{\min},1)}$ yields that at least two of $|\check{f}_{8,i-1}(x-\delta e_i)-f(x)|$, $|\check{f}_{8,i-1}(x)-f(x)|$ and $|\check{f}_{8,i-1}(x+\delta e_i)|-f(x)$ are smaller than

$$c_{100}(LW)^{-2\tilde{s}/d}+i\|f\|_{\mathcal{C}^s}\delta^{\min(s_{\min},1)}.$$

This implies that at least two of $\check{f}_{8,i}(x-\delta e_i)$, $\check{f}_{8,i}(x)$ and $\check{f}_{8,i}(x+\delta e_i)$ are in the interval

$$\left[f(x)-c_{100}(LW)^{-2\tilde{s}/d}-i\|f\|_{\mathcal{C}^s}\delta^{\min(s_{\min},1)},f(x)+c_{100}(LW)^{-2\tilde{s}/d}+i\|f\|_{\mathcal{C}^s}\delta^{\min(s_{\min},1)}\right].$$

Therefore, middle value $\check{f}_{8,i}(x)=\text{mid}\left(\check{f}_{8,i-1}(x-\delta e_i),\check{f}_{8,i}(x),\check{f}_{8,i-1}(x+\delta e_i)\right)$ belongs to the above interval, i.e.

$$\left|\check{f}_{8,i}(x)-f(x)\right|\leq c_{100}(LW)^{-2\tilde{s}/d}+i\|f\|_{\mathcal{C}^s}\delta^{\min(s_{\min},1)}.$$

Hence, the case for i holds. This completes the proof of the claim (S.39) by induction. In particularly, let $i=d$ and it follows

$$|\check{f}_{8,d}(x)-f(x)|\leq c_{100}(LW)^{-2\tilde{s}/d}+d\|f\|_{\mathcal{C}^s}\delta^{\min(s_{\min},1)},$$

for $x\in\Omega$. Since the exact value of δ has not yet been determined, we take it small enough to $\delta^{\min(s_{\min},1)}\leq(LW)^{-2\tilde{s}/d}$. Hence,

$$|\check{f}_{8,d}(x)-f(x)|\leq c_{100}(LW)^{-2\tilde{s}/d}+d\|f\|_{\mathcal{C}^s}\delta^{\min(s_{\min},1)}\leq c_{101}(LW)^{-2\tilde{s}/d},$$

for $x\in\Omega$ and c_{101} which is a constant independent of L and W. The recursive construction of the neural network makes the size verification simple, and the desired result follows.

□

Proof of Theorem S.3. By the proof of Theorem 6.2 in Jiao et al. (2023), we know that for any $f\in\mathcal{C}^s(\Omega)$ satisfying $\|f\|_\infty\leq B_1$ and any $L,W\geq 3$, there is some ReLU neural

network function $f_{\mathcal{FNN}} \in \mathcal{FNN}\left(d, c_{102} L \log L, c_{103} W \log W, B_1\right)$ satisfying

$$|f_{\mathcal{FNN}}(x) - f(x)| \leq c_{104} \left(LW\right)^{-2s/d_{\mathcal{M}}},$$

for every $x \in \mathcal{M}$. Here, c_{102} and c_{103} are constants independent of L, W and f. Combining the above conclusion and Corollary 2.2, the desired result follows immediately.

□

Proof of Theorem S.4. This proof is similar to the proof of Theorem 3.2. We first recall some notations. Let $s_{\max} = \max_{1\leq j\leq d} s_j$, $s_{\min} = \min_{1\leq j\leq d} s_j$, $s_i' = s_{\min}/s_i$ and $\tilde{s}$ is the harmonic mean of s_i for $1 \leq i \leq d$. Denote

$$\boldsymbol{k}(j) = \left(2^{\lfloor j s_1' \rfloor}, 2^{\lfloor j s_2' \rfloor}, ..., 2^{\lfloor j s_d' \rfloor}\right),$$

for each positive integer j. Given the sample sizes n and m, let $J = \lfloor \log_2(nm) \cdot \tilde{s}/(2\tilde{s}s_{\min} + ds_{\min}) \rfloor$ and c_{105} be a constant determined next. Then we define a multi-index set

$$\mathcal{V} = \left\{ \boldsymbol{v} = (v_1, v_2, ..., v_d) : -r+1 \leq v_i \leq 2^{\lfloor J s_i' \rfloor} \text{ and } v_i = rt + 1 \text{ for some } t \in \mathbb{Z} \right\}. \quad \text{(S.40)}$$

and the B-spline linear space

$$\left\{ \sum_{\boldsymbol{v}} a_{J,\boldsymbol{v}} 2^{-J s_{\min}} c_{105} \mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{v}} : a_{J,\boldsymbol{v}} \in \{0,1\}, \boldsymbol{v} \in \mathcal{V} \right\}. \quad \text{(S.41)}$$

We know that $|\mathcal{V}| = \prod_{i=1}^d (\lceil k_i/r \rceil + 1)^d \geq 2^{Jds_{\min}/\tilde{s}}/(2r)^d \geq (nm)^{d/(2\tilde{s}+d)} 2^{-d(s_{\min}+\tilde{s})/\tilde{s}} r^{-d}$ $= c_{106}(nm)^{d/(2\tilde{s}+d)}$, for a constant c_{106}. Consider $x' = x + te_i$. Note that

$$\begin{aligned}
&2^{-J s_{\min}} \frac{\left|\partial_i^{\lfloor s_i \rfloor} \mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{v}}(x) - \partial_i^{\lfloor s_i \rfloor} \mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{v}}(x')\right|}{\|x - x'\|^{s_i - \lfloor s_i \rfloor}} \\
=&2^{-J s_{\min}} k_i^{s_i} \frac{\left|\partial_i^{\lfloor s_i \rfloor} \mathcal{B}^d_{1,\boldsymbol{0}}(\boldsymbol{k}(J)x - \boldsymbol{v}) - \partial_i^{\lfloor s_i \rfloor} \mathcal{B}^d_{1,\boldsymbol{0}}(\boldsymbol{k}(J)x' - \boldsymbol{v})\right|}{k_i^{s_i - \lfloor s_i \rfloor} \|x - x'\|^{s_i - \lfloor s_i \rfloor}} \\
\leq& \frac{\left|\partial_i^{\lfloor s_i \rfloor} \mathcal{B}^d_{1,\boldsymbol{0}}(\boldsymbol{k}(J)x - \boldsymbol{v}) - \partial_i^{\lfloor s_i \rfloor} \mathcal{B}^d_{1,\boldsymbol{0}}(\boldsymbol{k}(J)x' - \boldsymbol{v})\right|}{k_i^{s_i - \lfloor s_i \rfloor} \|x - x'\|^{s_i - \lfloor s_i \rfloor}}.
\end{aligned}$$

It implies that

$$\|2^{-Js_{\min}}\mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{v}}\|_{\mathcal{C}^s} \le \|\mathcal{B}^d_{1,\boldsymbol{0}}\|_{\mathcal{C}^s}.$$

Recall that the support sets of each B-spline disjoint. Therefore, we have

$$\|\sum_{\boldsymbol{v}} a_{J,\boldsymbol{v}} 2^{-Js_{\min}} c_{105}\mathcal{B}^d_{\boldsymbol{k}(J),\boldsymbol{v}}\|_{\mathcal{C}^s} \le 2c_{105}\|\mathcal{B}^d_{1,\boldsymbol{0}}\|_{\mathcal{C}^s} = c_{105}c_{107} < \infty.$$

for any $\boldsymbol{a} = (a_{J,\boldsymbol{v}} : v \in \mathcal{V})$. In addition, we have

$$c_{108}c_{105}^2 \operatorname{Ham}(\boldsymbol{a},\boldsymbol{a}')2^{-2Js_{\min}-Jds_{\min}/\tilde{s}} \le \|f_{\boldsymbol{a}} - f_{\boldsymbol{a}'}\|^2_{L^2(\Omega)} \le c_{109}c_{105}^2 \operatorname{Ham}(\boldsymbol{a},\boldsymbol{a}')2^{-2Js_{\min}-Jds_{\min}/\tilde{s}},$$

where c_{108} and c_{109} are constants independent of n and m. We still use the Varshamov-Gilbert bound, and know that there exists a subset $\mathcal{A} \subseteq \{0,1\}^{|\mathcal{V}|}$ which satisfies $|\mathcal{A}| \ge \exp\{|\mathcal{V}|/8\}$, and $\operatorname{Ham}(\boldsymbol{a},\boldsymbol{a}') \ge |\mathcal{V}|/8$ for $\boldsymbol{a},\boldsymbol{a}' \in \mathcal{A}, \boldsymbol{a} \ne \boldsymbol{a}'$.

Similar with the previous proof, to show the $(nm)^{-2\tilde{s}/(\tilde{s}+d)}$ term in the lower bound, we assume that Z is a Gaussian process and the measurement noise ϵ is normal random variables with variance B_3. Hence, using the KL divergence of two Gaussian probability measures, we have

$$\begin{aligned}
\operatorname{KL}(\mathcal{P}_{f_{\boldsymbol{a}}}, \mathcal{P}_{f_{\boldsymbol{a}'}}) =& \mathbb{E}\left[(\mu_{X,f_{\boldsymbol{a}}} - \mu_{X,f_{\boldsymbol{a}'}})C_X^{-1}(\mu_{X,f_{\boldsymbol{a}}} - \mu_{X,f_{\boldsymbol{a}'}})\right] \\
\le& nmB_3^{-1}\|f_{\boldsymbol{a}} - f_{\boldsymbol{a}'}\|^2_{L^2(\Omega)} \\
\le& c_{109}c_{105}^2 nmB_3^{-1}2^{-2Js_{\min}-Jds_{\min}/\tilde{s}}|\mathcal{V}| \\
\le& c_{110}c_{105}^2 B_3^{-1}(nm)^{d/(2\tilde{s}+d)},
\end{aligned}$$

where $\mu_{X,f_{\boldsymbol{a}}}$, $\mu_{X,f_{\boldsymbol{a}'}}$ and C_X are given by (S.28) and (S.29), and c_{110} is a constant independent of n and m. We can also check that

$$\begin{aligned}
\|f_{\boldsymbol{a}} - f_{\boldsymbol{a}'}\|^2_{L^2(\Omega)} \ge& c_{108}c_{105}^2 \operatorname{Ham}(\boldsymbol{a},\boldsymbol{a}')2^{-2Js-Jd} \\
\ge& c_{111}c_{105}^2(nm)^{-2\tilde{s}/(2\tilde{s}+d)},
\end{aligned}$$

with $\boldsymbol{a}, \boldsymbol{a}' \in \mathcal{A}$ and $\boldsymbol{a} \neq \boldsymbol{a}'$. Here c_{111} is a constant independent of sample size. Then it follows from Fano's inequality that

$$\begin{aligned}\max_{a\in\mathcal{A}} \mathbb{E}_{f_{\boldsymbol{a}}}\left[\int_\Omega (\tilde{f}-f_{\boldsymbol{a}})^2 d\mathcal{P}_X\right] &= \max_{a\in\mathcal{A}} \mathbb{E}_{f_{\boldsymbol{a}}}\left[\left\|\tilde{f}-f_{\boldsymbol{a}}\right\|_{L^2(\Omega)}^2\right] \\ \geq \frac{c_{111}c_{105}^2(nm)^{-2s/(2s+d)}}{4}&\left(1-\frac{c_{110}c_{105}^2B_3^{-1}(nm)^{d/(2\tilde{s}+d)}+\log 2}{2^{-3}c_{106}(nm)^{d/(2\tilde{s}+d)}}\right),\end{aligned}$$

for any estimator $\tilde{f}$. Let c_{105} small enough such that $c_{105}c_{107} \leq 1$ and $c_{110}c_{105}^2B_3^{-1} \leq 2^{-5}c_{106}$. Then

$$1-\frac{c_{110}c_{105}^2B_3^{-1}(nm)^{d/(2\tilde{s}+d)}+\log 2}{2^{-3}c_{106}(nm)^{d/(2\tilde{s}+d)}} \geq \frac{1}{2},$$

when nm is larger than some fixed number. Therefore, there exists a constant c_{112} free of n and m such that

$$\begin{aligned}\inf_{\tilde{f}}\sup_{\mathcal{P}_{Z,\epsilon}\in\mathcal{Q}_{Z,\epsilon}} \mathbb{E}_{f^\circ}\left[\int_\Omega (\tilde{f}-f^\circ)^2 d\mathcal{P}_X\right] &\geq \inf_{\tilde{f}}\max_{a\in\mathcal{A}} \mathbb{E}_{f_{\boldsymbol{a}}}\left[\int_\Omega (\tilde{f}-f_{\boldsymbol{a}})^2 d\mathcal{P}_X\right] \\ &\geq c_{113}(nm)^{-2s/(2s+d)}.\end{aligned}$$

The infimum here is correct because Fano inequality holds for any estimator. Moreover, after the same argument as (S.31), we have

$$\inf_{\tilde{f}}\sup_{\mathcal{P}_{Z,\epsilon}\in\mathcal{Q}_{Z,\epsilon}} \mathbb{E}_{f^\circ}\left[\int_\Omega (\tilde{f}-f^\circ)^2 d\mathcal{P}_X\right] \geq B_2 n^{-1}.$$

Then the desired result follows. □

Proof of Theorem S.1. Using Lemma S.11, the proof is given by the construction of partition $\mathcal{A}$ in (ii-b) of the proof of Theorem 5 in Suzuki & Nitanda (2021) with $\tilde{d}=1$. □

S.6 Proofs of Lemmas

Proof of Lemma S.2. For the purpose of bounding $\int_\Omega (\hat{f}-f^\circ)^2 d\mathcal{P}_X$ with its empirical version sharply, we use the peeling technique combined with Talagrand's inequality. Assume

that r is a positive number. We first consider

$$\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\int_{\Omega}(\hat{f}-f^{\circ})^{2}d\mathcal{P}_{X}-\left(\hat{f}\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)^{2}}{\int_{\Omega}(\hat{f}-f^{\circ})^{2}d\mathcal{P}_{X}+r}\right|,$$

which is bounded by

$$\sup_{f\in\mathcal{F}}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\int_{\Omega}(f-f^{\circ})^{2}d\mathcal{P}_{X}-\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)^{2}}{\int_{\Omega}(f-f^{\circ})^{2}d\mathcal{P}_{X}+r}\right|.$$

As a result of the symmetry and convexity, we have

$$\begin{aligned}&\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\int_{\Omega}(f-f^{\circ})^{2}d\mathcal{P}_{X}-\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)^{2}}{\int_{\Omega}(f-f^{\circ})^{2}d\mathcal{P}_{X}+r}\right|\right]\\ \leq&2\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\sigma_{ij}\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)^{2}}{\int_{\Omega}(f-f^{\circ})^{2}d\mathcal{P}_{X}+r}\right|\right],\end{aligned}$$

which is usually called as the symmetrization conclusion of Rademacher process. Let the real number r be greater than r^{*}. Note that $\mathcal{F}$ is the union of $\mathcal{F}(r)$ and $\mathcal{F}(4^{k}r)\backslash\mathcal{F}(4^{k-1}r)$ for $k\geq 1$, we have

$$\begin{aligned}&\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\sigma_{ij}\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)^{2}}{\int_{\Omega}(f-f^{\circ})^{2}d\mathcal{P}_{X}+r}\right|\right]\\ \leq&\mathbb{E}\left[\sup_{f\in\mathcal{F}(r)}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\sigma_{ij}\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)^{2}}{\int_{\Omega}(f-f^{\circ})^{2}d\mathcal{P}_{X}+r}\right|\right]\\ &+\sum_{k=1}^{\infty}\mathbb{E}\left[\sup_{f\in\mathcal{F}(4^{k}r)\backslash\mathcal{F}(4^{k-1}r)}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\sigma_{ij}\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)^{2}}{\int_{\Omega}(f-f^{\circ})^{2}d\mathcal{P}_{X}+r}\right|\right],\end{aligned}$$

which could be further bounded by

$$\begin{aligned}&\frac{1}{r}\mathbb{E}\left[\sup_{f\in\mathcal{F}(r)}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\sigma_{ij}\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)^{2}\right|\right]\\ &+\sum_{k=1}^{\infty}\frac{1}{(4^{k-1}+1)r}\mathbb{E}\left[\sup_{f\in\mathcal{F}(4^{k}r)}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\sigma_{ij}\left(f\left(X_{ij}\right)-f^{\circ}\left(X_{ij}\right)\right)^{2}\right|\right].\end{aligned}$$

The last inequality stems from the definition of $\mathcal{F}(\cdot)$. As a reminder, the functions in $\mathcal{F}$ are bounded in infinity norm. By invoking Lemma S.7, we can further upper-bound the

Rademacher complexity of $\{(f-f^\circ)^2 : f \in \mathcal{F}(r')\}$ with

$$\begin{aligned}
&\mathbb{E}\left[\sup_{f\in\mathcal{F}(r')}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\sigma_{ij}\left(f\left(X_{ij}\right)-f^\circ\left(X_{ij}\right)\right)^2\right|\right]\\
\leq& 4B_1\mathbb{E}\left[\sup_{f\in\mathcal{F}(r')}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\sigma_{ij}\left(f\left(X_{ij}\right)-f^\circ\left(X_{ij}\right)\right)\right|\right]\\
\leq& 4B_1\phi(r'),
\end{aligned}$$

for any $r' \geq r^*$. Hence, by substituting this bound into the last line of the earlier formula, we obtain

$$\frac{4B_1\phi(r)}{r}+\sum_{k=1}^{\infty}\frac{4B_1\phi(4^k r)}{(4^{k-1}+1)r}\leq\frac{4B_1\phi(r)}{r}+\sum_{k=1}^{\infty}\frac{2^{k+2}B_1\phi(r)}{(4^{k-1}+1)r}\leq\frac{16B_1\phi(r)}{r},$$

where the first inequality is because that ϕ is sub-root and the second inequality is derived by simple algebra. Combining the above several results yields

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X-\left(f\left(X_{ij}\right)-f^\circ\left(X_{ij}\right)\right)^2}{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X+r}\right|\right]\leq\frac{32B_1\phi(r)}{r}. \tag{S.42}$$

Then, to apply Talagrand's inequality, we shall verify the remaining conditions. It is easy to show

$$\mathbb{E}\left[\frac{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X-\left(f\left(X\right)-f^\circ\left(X\right)\right)^2}{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X+r}\right]=0, \tag{S.43}$$

$$\left|\frac{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X-\left(f\left(X\right)-f^\circ\left(X\right)\right)^2}{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X+r}\right|\leq\frac{4B_1}{r}, \tag{S.44}$$

and

$$\mathbb{E}\left[\left|\frac{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X-\left(f\left(X\right)-f^\circ\left(X\right)\right)^2}{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X+r}\right|^2\right]\leq\frac{\mathbb{E}\left[\left(f\left(X\right)-f^\circ\left(X\right)\right)^4\right]}{4r\int_\Omega(f-f^\circ)^2d\mathcal{P}_X}\leq\frac{B_1^2}{r}. \tag{S.45}$$

The first inequality of (S.45) is because both that the expectation of the numerator is zero and using AM–GM inequality for the denominator. According, we can use Talagrand's concentration inequality now. By Lemma S.8 and (S.42), (S.43), (S.44), (S.45), with prob-

ability at least $1-\exp\{-t\}$, it holds

$$\begin{aligned}
&\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\int_\Omega(\hat{f}-f^\circ)^2d\mathcal{P}_X-\left(\hat{f}\left(X_{ij}\right)-f^\circ\left(X_{ij}\right)\right)^2}{\int_\Omega(\hat{f}-f^\circ)^2d\mathcal{P}_X+r}\\
\leq&\sup_{f\in\mathcal{F}}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X-\left(f\left(X_{ij}\right)-f^\circ\left(X_{ij}\right)\right)^2}{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X+r}\right| \qquad \text{(S.46)}\\
\leq&\frac{64B_1\phi(r)}{r}+\sqrt{\frac{8B_1^2t}{nmr}}+\frac{140B_1^2t}{nmr}.
\end{aligned}$$

Therefore, there are universal constants c_{38} and c_{39} such that $r_1(t)=2c_{38}B_1^2r^*+\frac{2c_{39}B_1^2t}{nm}$ satisfies

$$\frac{64B_1\phi(r_1(t))}{r_1(t)}+\sqrt{\frac{8B_1^2t}{nmr_1(t)}}+\frac{140B_1^2t}{nmr_1(t)}\leq\frac{64B_1\phi(2c_{38}B_1^2r^*)}{2c_{38}B_1^2r^*}+\sqrt{\frac{4}{c_{39}}}+\frac{70}{c_{39}}\leq\frac{1}{2}.$$

Substituting $r=r_1(t)$ into (S.46), the desired result follows. □

Proof of Lemma S.3. We use a similar method with Lemma S.2 to consider

$$\left|\frac{\left(\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}-\mathbb{E}\left[\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}\right]\right)\left(\hat{f}\left(X_{ij}\right)-f^\circ\left(X_{ij}\right)\right)}{\int_\Omega(\hat{f}-f^\circ)^2d\mathcal{P}_X+r}\right|,$$

which is bounded by

$$\sup_{f\in\mathcal{F}}\left|\frac{\left(\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}-\mathbb{E}\left[\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}\right]\right)\left(f\left(X_{ij}\right)-f^\circ\left(X_{ij}\right)\right)}{\int_\Omega(f-f^\circ)^2d\mathcal{P}_X+r}\right|.$$

After some discussion similar to the proof of Lemma S.2, we obtain, with probability at least $1-\exp\{-t\}$,

$$\begin{aligned}
&\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\left(\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}-\mathbb{E}\left[\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}\right]\right)\left(\hat{f}\left(X_{ij}\right)-f^\circ\left(X_{ij}\right)\right)}{\int_\Omega(\hat{f}-f^\circ)^2d\mathcal{P}_X+r}\right|\\
\leq&\sup_{f\in\mathcal{F}}\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\frac{\left(\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}-\mathbb{E}\left[\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}\right]\right)\left(\hat{f}\left(X_{ij}\right)-f^\circ\left(X_{ij}\right)\right)}{\int_\Omega(\hat{f}-f^\circ)^2d\mathcal{P}_X+r}\right|\\
\leq&\frac{32\beta_\epsilon\phi(r)}{r}+\sqrt{\frac{2B_3^2t}{nmr}}+\frac{140B_1\beta_\epsilon t}{nmr}.
\end{aligned}$$

The second term on the right side is different from its counterpart in step 3 of the proof of Theorem 2.1 because the variance estimation is smaller due to the independent of X_{ij} and

ϵ_{ij}. Choosing constants c_{114} and c_{115} such that $r_3(t) = c_{114}\beta_\epsilon^2 r^* + \frac{c_{115}(B_1\beta_\epsilon + B_3^2)t}{nm}$ satisfy

$$\frac{32\beta_\epsilon\phi(r)}{r} + \sqrt{\frac{2B_3^2 t}{nmr}} + \frac{140B_1\beta_\epsilon t}{nmr} \leq \frac{1}{16}.$$

Therefore, with probability at least $1 - \exp\{-t\}$,

$$\left|\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m}\left(\mathcal{T}_{\beta_\epsilon}\epsilon_{ij} - \mathbb{E}\left[\mathcal{T}_{\beta_\epsilon}\epsilon_{ij}\right]\right)\left(\hat{f}\left(X_{ij}\right) - f^\circ\left(X_{ij}\right)\right)\right|$$
$$\leq \frac{1}{16}\int_\Omega (\hat{f} - f^\circ)^2 d\mathcal{P}_X + \frac{c_{114}\beta_\epsilon^2 r^*}{16} + \frac{c_{115}(B_1\beta_\epsilon + B_3^2)t}{16nm}.$$

This completes the proof. □

Proof of Lemma S.4. The main proof of Step 1 can be found in Lu et al. (2021).

Step 1. We first consider the case $K = L^2W^2$. Applying Lemma 2.2 in Shen et al. (2019) or Lemma 5.2 in Lu et al. (2021) yields that there is a ReLU neural network

$$\check{f}_1 \in \mathcal{FNN}\left(\text{inputdim} = 1, \text{widthvec} = [2W; 4WL - 1]\right)$$

satisfying

$$\check{f}_1\left(\frac{k}{LW^2}\right) = \check{f}_1\left(\frac{k+1}{LW^2} - \delta\right) = k$$

for $k = 0, 1, ..., LW^2 - 2$,

$$\check{f}_1\left(\frac{LW^2 - 1}{LW^2}\right) = \check{f}_1(1) = LW^2 - 1, \quad \check{f}_1(2) = 0,$$

and being linear on each interval between the nodes above. These imply that

$$\check{f}_1(x) = k, \quad \text{when } x \in \left[\frac{k}{LW^2}, \frac{k+1}{LW^2} - \delta \cdot I_{\{k \leq LW^2 - 2\}}\right],$$

for $k = 0, 1, ..., LW^2 - 2$, and

$$\check{f}_1(x) = LW^2 - 1, \quad \text{when } x \in \left[\frac{LW^2 - 1}{LW^2}, 1\right].$$

To represent the shallow and wide neural network $\check{f}_1$ with a deep and narrow one, we use Proposition 2.2 in Shen (2020) and get there is a neural network

$$\check{f}_2 \in \mathcal{FNN}(1, 2L + 1, 4W + 2)$$

such that $\check{f}_2(x)$ is always equal to $\check{f}_1(x)$.

Reconsider the construction of $\check{f}_2(x)$ and set $W = 1$, we know there is a neural network

$$\check{f}_3 \in \mathcal{FNN}\left(1, 2L+1, 6\right)$$

such that

$$\check{f}_3\left(x\right) = k, \quad \text{when } x \in \left[\frac{k}{L}, \frac{k+1}{L} - \delta \cdot I_{\{k \leq L-2\}}\right]$$

for $k = 0, 1, ..., L-2$, and

$$\check{f}_3\left(x\right) = L-1, \quad \text{when } x \in \left[\frac{L-1}{L}, 1\right].$$

After constructing $\check{f}_2$ and $\check{f}_3$, we analyze the original problem. Let

$$\check{f}_4 = L\check{f}_2(x) + \check{f}_3\left(LW^2(x - \check{f}_2(x))\right),$$

it is easy to verify

$$\check{f}_4(x) = k, \quad \text{when } x \in \left[\frac{k}{L^2W^2}, \frac{k+1}{L^2W^2} - \delta \cdot I_{\{k \leq L^2W^2-2\}}\right],$$

for $k = 0, 1, \cdots, L^2W^2 - 1$. And now we should only specify the width and depth of the desired network. To this goal, we decompose the function $\check{f}_4$. The first composited network inputs x and outputs $(\check{f}_2(x), x)$. Then the second composited network inputs $(\check{f}_2(x), x)$ and outputs $(\check{f}_2(x), L^2W(x - \check{f}_2(x))$. Using the results of the previous layer, the last composited network calculates $\check{f}_4 = L\check{f}_2(x) + \check{f}_3\left(L^2W(x - \check{f}_2(x))\right)$. This yields the width of $\check{f}_4$ is less than $4W + 2 + 1 \leq 4W + 3$ and depth $\leq 2L + 1 + 2 + 2L + 1 + 2 = 4L + 6$. This completes the proof for $K = L^2W^2$.

Step 2. When $K < L^2W^2$, Let

$$\check{f}_5(x) = \check{f}_4'\left(\frac{Kx}{L^2W^2}\right).$$

The construction of $\check{f}_4'$ uses the previous step, and the only difference is that δ' is taken as $\frac{K\delta}{L^2W^2}$. One can check that $\check{f}_5$ satisfies the requirements. The desired result follows. □

Proof of Lemma S.6. Step 1. We first consider the case $d = 1$. Applying Lemma 5.3 in Lu et al. (2021), we know there is a ReLU neural network

$$\check{f}_1 \in \mathcal{FNN}\left(1, 7(r-1)(r-2)L, 9W + r + 7\right)$$

satisfying

$$\left|\check{f}_1(x) - x^{r-1}\right| \leq 9(r-2)(W+1)^{-7(r-1)L},$$

for $x \in [0,1]$. Let $\check{f}_2 = r^{r-1}\check{f}_1\left(\frac{x}{r}\right)$. Then

$$\left|\check{f}_2(x) - x^{r-1}\right| \leq 9(r-2)r^{r-1}(W+1)^{-7(r-1)L},$$

for $x \in [0,r]$. Using $\check{f}_2$ and recalling (S.1), let

$$\check{f}_3(x) = \frac{1}{(r-1)!}\sum_{i=0}^{r}(-1)^i\binom{r}{i}\check{f}_2 \circ \mathrm{ReLU}(x-i).$$

In order to make the composition in the next step more convenient, we truncate the function

$$\check{f}_4(x) = \mathrm{ReLU}\left(\check{f}_3(x)\right) - \mathrm{ReLU}\left(\check{f}_3(x) - 1\right).$$

Then the approximation error is bounded by

$$\begin{aligned}
\left|\check{f}_4(x) - B_r(x)\right| \leq & \left|\check{f}_3(x) - B_r(x)\right| \\
\leq & \frac{1}{(r-1)!}\sum_{i=0}^{r}\binom{r}{i}\left|\check{f}_2 \circ \mathrm{ReLU}(x-i) - (x-i)_+^{r-1}\right| \\
< & \frac{9 \cdot 2^r r^r}{(r-1)!}(W+1)^{-7(r-1)L}.
\end{aligned}$$

Step 2. Now we focus on the multivariate case for $\mathcal{B}^d_{\boldsymbol{k},\mathbf{0}}(x)$ with $x = (x_1, x_2, ..., x_d)$. By Lemma 5.3 in Lu et al. (2021), we know there is a ReLU neural network

$$\check{f}_5 \in \mathcal{FNN}\left(d, 7d(d-1)L, 9W + d + 8\right)$$

satisfying

$$\left|\check{f}_5(x_1, x_2, ..., x_d) - x_1 x_2 \cdots x_d\right| \leq 9(d-1)(N+1)^{-7dL},$$

for $x_i \in [0,1]$. Let

$$\check{f}_6(x) = \check{f}_5\left(\check{f}_4(x_1), \check{f}_4(x_2), ..., \check{f}_4(x_d)\right).$$

Hence, we have

$$\begin{aligned}
&\left|\check{f}_6(x) - \mathcal{B}^d_{1,\mathbf{0}}(x)\right| \\
=&\left|\check{f}_5\left(\check{f}_4(x_1), \check{f}_4(x_2), ..., \check{f}_4(x_d)\right) - \prod_{i=1}^d B_r(x_i)\right| \\
\leq&\left|\check{f}_5\left(\check{f}_4(x_1), \check{f}_4(x_2), ..., \check{f}_4(x_d)\right) - \prod_{i=1}^d \check{f}_4(x_i)\right| + \left|\prod_{i=1}^d \check{f}_4(x_i) - \prod_{i=1}^d B_r(x_i)\right|.
\end{aligned}$$

By the construction of $\check{f}_5$, the first term is bounded by $9(d-1)(W+1)^{-7dL}$. Then the second term is bounded by

$$\begin{aligned}
&\left|\prod_{i=1}^d \check{f}_4(x_i) - \prod_{i=1}^d B_r(x_i)\right| \\
\leq&\sum_{j=0}^{d-1}\left|\prod_{i=1}^{d-j} \check{f}_4(x_i) \prod_{i=d-j+1}^{d} B_r(x_i) - \prod_{i=1}^{d-j-1} \check{f}_4(x_i) \prod_{i=d-j}^{d} B_r(x_i)\right| \\
\leq&\sum_{j=0}^{d-1}\left|\check{f}_4(x_{d-j}) - B_r(x_{d-j})\right| \\
\leq&\frac{9\cdot 2^r r^r d}{(r-1)!}(W+1)^{-7(r-1)L}.
\end{aligned}$$

The first inequality is given by triangle inequality, the second inequality is due to $|\check{f}_4|, |B_r| \leq 1$, and the last inequality is implied from the approximation error of $\check{f}_4$. Hence, it follows that the approximation error satisfies the requirement. Finally, it is easy to check the size of neural network $\check{f}_6$. This completes the proof. □

Proof of Lemma S.11. The proof is divided into two parts, each proving a term on the right-hand side. The proof of the nm terms comes from Theorem 1 in Zhang et al. (2002), and the n terms are the same as the other lower bound in this paper. To prove the nm term, we assume Z is a Gaussian process and ϵ is a Gaussian noise with variance B_3. For simplicity, we write

$$\varphi_{ij}(x) = \varphi_{ij}(x, X_{11}, \ldots, X_{nm}).$$

By Zhang et al. (2002), we know that $P(\mathcal{D}) \geq 1 - o(1)$. Denote $I_{\mathcal{D}}$ as indicator function of event $\mathcal{D}$. Recalling the definition of linear estimator $\tilde{f}_L$, we have

$$\begin{aligned}
\mathcal{E}_{f^\circ}(\tilde{f}_L) =& \mathbb{E}\left[\left\|\sum_{i=1}^{n}\sum_{j=1}^{m} Y_{ij}\varphi_{ij}(x) - f^\circ(x)\right\|_{L^2(\Omega)}^2\right] \\
\geq& \mathbb{E}\left[\left\|\sum_{i=1}^{n}\sum_{j=1}^{m} f^\circ(X_{ij})\varphi_{ij}(x) - f^\circ(x)\right\|_{L^2(\Omega)}^2 I_{\mathcal{D}}\right] \\
&+ B_3 \sum_{i=1}^{n}\sum_{j=1}^{m} \mathbb{E}\left[\|\varphi_{ij}(x)\|_{L^2(\Omega)}^2 I_{\mathcal{D}}\right].
\end{aligned} \tag{S.47}$$

Because $\mathcal{A}$ is a partition of Ω and $|\mathcal{A}| \geq {c_{116}}^{-1}2^K$ for some constant c_{116}, there exists $A \in \mathcal{A}$ such that

$$\sum_{i=1}^{n}\sum_{j=1}^{m} \mathbb{E}\left[\|\varphi_{ij}(x)\|_{L^2(A)}^2 I_{\mathcal{D}}\right] \leq c_{116}2^{-K}B_3^{-1}\mathcal{E}_{f^\circ}(\tilde{f}_L). \tag{S.48}$$

Fix A and use condition (i), there is a function $g \in \mathcal{F}^\circ$ satisfying $g(x) \geq \frac{1}{2}b_5 b_6$ for all $x \in A$, which further implies

$$\mathbb{E}\left[\|g(x)\|_{L^2(A)} I_{\mathcal{D}}\right] \geq \frac{1}{2}b_5 b_6 2^{-K/2}P(\mathcal{D}). \tag{S.49}$$

Denote the random function

$$h(x) = \sum_{i=1}^{n}\sum_{j=1}^{m} g(X_{ij})\varphi_{ij}(x).$$

Then, we have

$$\begin{aligned}
\mathbb{E}\left[\|h(x)\|_{L^2(\Omega)}^2 I_{\mathcal{D}}\right] =& \mathbb{E}\left[\int_\Omega \left(\sum_{i=1}^{n}\sum_{j=1}^{m} g(X_{ij})\varphi_{ij}(x)\right)^2 dx \cdot I_{\mathcal{D}}\right] \\
\leq& \mathbb{E}\left[\left(\sum_{i=1}^{n}\sum_{j=1}^{m} g^2(X_{ij})\right)\int_\Omega \sum_{i=1}^{n}\sum_{j=1}^{m}\varphi_{ij}^2(x)\,dx \cdot I_{\mathcal{D}}\right] \\
\leq& c_{116}2^{-(K'+K)}nmb_7 b_5^2 B_3^{-1}\mathcal{E}_{f^\circ}(\tilde{f}_L),
\end{aligned}$$

where the first inequality is given by Cauchy inequality and the second inequality is due to (S.48) and condition (ii). This implies

$$\mathbb{E}\left[\|h(x)\|_{L^2(\Omega)} I_{\mathcal{D}}\right] \leq \sqrt{\mathbb{E}\left[\|h(x)\|_{L^2(\Omega)}^2 I_{\mathcal{D}}\right]} \leq \sqrt{c_{116}2^{-(K'+K)}nmb_7 b_5^2 B_3^{-1}\mathcal{E}_{f^\circ}(\tilde{f}_L)}. \tag{S.50}$$

$$\begin{aligned}\sup_{\mathcal{P}_{Z,\epsilon}\in\mathcal{Q}_{Z,\epsilon}}\mathcal{E}_{f^\circ}(\tilde{f}_L) \geq& \mathbb{E}\left[\|h(x)-g(x)\|^2_{L^2(\Omega)} I_{\mathcal{D}}\right]\\ \geq& \mathbb{E}\left[\|h(x)-g(x)\|^2_{L^2(A)} I_{\mathcal{D}}\right]\\ \geq& \mathbb{E}\left[\left(\|g(x)\|_{L^2(A)} I_{\mathcal{D}} - \|h(x)\|_{L^2(A)} I_{\mathcal{D}}\right)^2\right]\\ \geq& \left(\mathbb{E}\left[\|g(x)\|_{L^2(A)} I_{\mathcal{D}}\right] - \mathbb{E}\left[\|h(x)\|_{L^2(A)} I_{\mathcal{D}}\right]\right)^2\end{aligned}$$

where the first inequality is implied from (S.47), the third inequality is due to the Triangle inequality, and the last inequality is given by Jensen inequality. Recalling (S.49) and (S.50), we have

$$\begin{aligned}&\mathbb{E}\left[\|g(x)\|_{L^2(A)} I_{\mathcal{D}}\right] - \mathbb{E}\left[\|h(x)\|_{L^2(A)} I_{\mathcal{D}}\right]\\ &\quad\geq \frac{1}{2}b_5 b_6 2^{-K/2}P(\mathcal{D}) - \sqrt{c_{116}2^{-(K'+K)}nmb_7 b_5^2 B_3^{-1}\mathcal{E}_{f^\circ}(\tilde{f}_L)}.\end{aligned} \tag{S.51}$$

If the right side of (S.51) $\leq \frac{1}{4}b_5 b_6 2^{-K/2}P(\mathcal{D})$, i.e.

$$\frac{1}{4}b_5 b_6 2^{-K/2}P(\mathcal{D}) \leq \sqrt{c_{116}2^{-(K'+K)}nmb_7 b_5^2 B_3^{-1}\mathcal{E}_{f^\circ}(\tilde{f}_L)},$$

we have

$$\mathcal{E}_{f^\circ}(\tilde{f}_L) \geq \frac{b_6^2 2^{K'}B_3 P(\mathcal{D})}{16c_{116}nmb_7}. \tag{S.52}$$

If the right side of (S.51) $\geq \frac{1}{4}b_5 b_6 2^{-K/2}P(\mathcal{D}) > 0$, we have

$$\begin{aligned}\sup_{\mathcal{P}_{Z,\epsilon}\in\mathcal{Q}_{Z,\epsilon}}\mathcal{E}_{f^\circ}(\tilde{f}_L) \geq& \left(\mathbb{E}\left[\|g(x)\|_{L^2(A)} I_{\mathcal{D}}\right] - \mathbb{E}\left[\|h(x)\|_{L^2(A)} I_{\mathcal{D}}\right]\right)^2\\ \geq& \frac{1}{16}b_5^2 b_6^2 2^{-K}P(\mathcal{D})^2.\end{aligned} \tag{S.53}$$

Combining the discussion in (S.52) and (S.53) and noting that $P(\mathcal{D}) \geq 1 - o(1)$, we have

$$\inf_{\tilde{f}_L}\sup_{\mathcal{P}_{Z,\epsilon}\in\mathcal{Q}_{Z,\epsilon}}\mathcal{E}_{f^\circ}(\tilde{f}_L) \gtrsim \min\left(\frac{b_6^2 2^{K'}}{b_7 nm} + b_5^2 b_6^2 2^{-K}\right). \tag{S.54}$$

Moreover, to show the remaining $1/n$ term in the lower bound, it now suffices to consider Z_i as constant functions over $x \in \Omega$ and the measurement noise ϵ_{ij} as zero. Assume that $\mathrm{Var}(Z_i(X)|X) = B_2$. Then the problem becomes the mean estimation from n i.i.d.

observations. It is well known that the rate

$$\inf_{\tilde{f}} \sup_{\mathcal{P}_{Z,\epsilon} \in \mathcal{Q}_{Z,\epsilon}} \mathcal{E}_{f^\circ}(\tilde{f}_L) \geq B_2 n^{-1}. \tag{S.55}$$

Then combining (S.54) and (S.55) completes the proof.

□

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